
Conformal Open-Vocabulary Detection: Hallucination FDR Control via Self-Normalized Betting e-Values

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Abstract

Open-vocabulary detectors return high-confidence boxes for categories that are not present: in a COCO absent-query pilot, the top-decile detections are 100% hallucinations. We show that, for category-name or fixed-template prompts whose absence labels are auditable, post-hoc false discovery rate (FDR) control is achievable on a frozen detector by applying e-BH with self-normalized conformal betting e-values within each image-query family, which tolerates the arbitrary dependence induced by shared proposals and NMS. The power of this procedure is governed by a closed-form rejection condition, which we call the *activation frontier*. It is necessary and sufficient, and is expressed in the family size, the calibration pool aggregate, and the target level. The rejection condition shows how family size, calibration evidence, and score separation determine whether e-BH makes discoveries, and it suggests practical variants such as score filtering, cluster-level aggregation, and template-specific calibration. On COCO and VOC with GroundingDINO, OWL-ViT, and YOLO-World, per-family FDP stays below the nominal level at $\alpha = 0.10$ across all six configurations (max 1.7%) with recall 9.7–40.9%; a stratified empirical Bernstein bound provides a non-vacuous pooled FDP certificate ($\varepsilon = 0.074$) on the 57% of families with $K \leq 5$; released detections on full COCO reach 93.6% pooled precision versus 72.9% for p-value BH; cluster-level aggregation lifts operational recall $1.7\times$ (COCO/GDino) and $5.8\times$ (RegionCLIP, no internal NMS). On LVIS and Open Images, observed failures are consistent with the contamination mechanism identified by the condition, which also indicates the bound under which validity is restored.

1 Introduction

A binary conformal e-value that tests whether a detector score exceeds a high calibration-null quantile is formally valid, yet in our COCO pilot it rejects *zero* detections despite the top-decile being 100% hallucinations (FDP = 1.0) across 2,992 image-query families. The pilot points to quantization rather than absence of signal: the e-value takes only two values, while e-BH needs evidence that scales with family size. Looking at the e-BH rejection condition directly, we call it the *activation frontier* (Theorem 1): in an image-query family of size K , e-BH rejects r candidates only if the r -th largest evidence exceeds $KC_m/[\alpha r(m+1) - K]$. The inequality is elementary but will be used below to analyze power separately from validity, and to trace three obstacles: detector-limited recall, non-exhaustive annotations, and the failure of the natural fix that clips scores to a high quantile.

Recent work documents the OVD hallucination problem [1], yet detection-side reliability tools still focus on coverage and recall rather than false discoveries. Standard Benjamini–Hochberg (BH) procedures [2] require independence or PRDS [3], assumptions that are difficult to justify after proposal sharing and NMS. Existing conformal detection tools [4, 5] do not target family-level

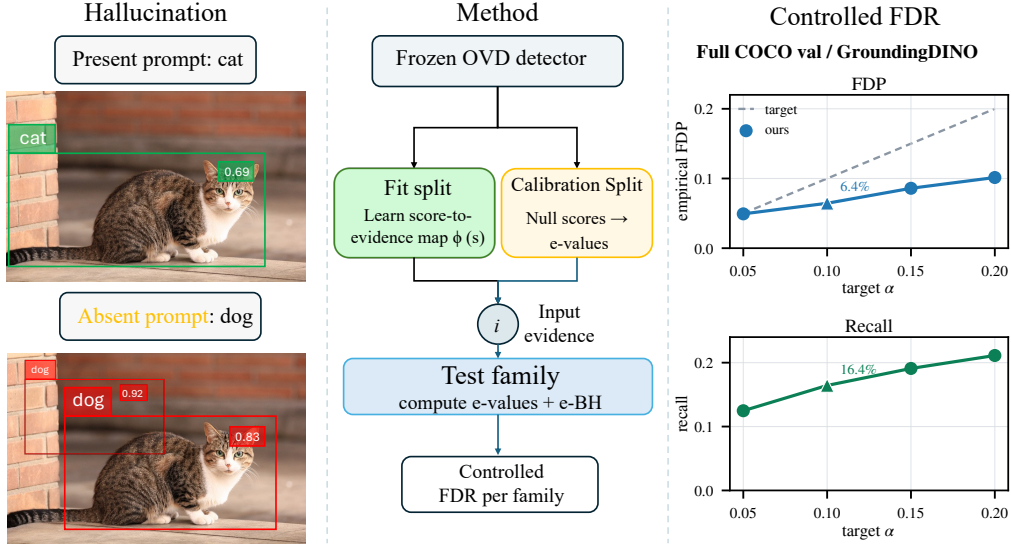


Figure 1: Overview: absent-prompt boxes can score highly; our frozen-detector wrapper converts scores to e-values and applies per-family e-BH to control FDR with non-trivial recall.

37 FDR. We therefore study post-hoc hallucination filtering as a finite-sample FDR-control problem for
 38 frozen OVD detectors, allowing arbitrary dependence among candidate boxes within each image-
 39 query family.

40 The method requires “absent in the annotation” to imply “truly absent”; this holds for exhaustive-
 41 per-class datasets (COCO, VOC), while non-exhaustive annotations in LVIS [6] and Open Im-
 42 ages [7] can sharply reduce power unless contamination is bounded or the evaluation vocabulary
 43 is curated (Theorem 3; Section 6). Our main experiments use category-name prompts and fixed
 44 template banks for which absent/present status can be audited; fully free-form language queries fall
 45 outside this setting and remain future work.

46 We treat the frozen detector together with its score filter and NMS stack as a deterministic candidate
 47 generator, and target finite-sample control of the hallucination false discovery rate (FDR) on released
 48 detections within each image-query family. The feasible-regime method has three steps. First, on a
 49 held-out fit split, we learn a nonnegative score-to-evidence map ϕ by matching candidates to ground
 50 truth and fitting a density-ratio style estimator. Second, on a null-only calibration split, we turn each
 51 test candidate score into a self-normalized conformal betting e-value. Its validity follows from a
 52 one-line exchangeability argument (Lemma 1) and does not require any concentration inequality or
 53 clipping. Third, we apply e-BH [8] within each image-query family, which yields valid FDR control
 54 under arbitrary dependence among candidates in that family.

55 **Contributions.** We formulate image-query hallucination filtering as finite-sample family-level
 56 FDR control under arbitrary within-family dependence and derive the e-BH rejection condition
 57 $T_r(K, C_m) = KC_m/[\alpha r(m+1) - K]$ that characterizes its power. Combined with the contami-
 58 nation analysis, this condition leads to a deployment-level distinction between exhaustive annota-
 59 tion, bounded contamination, and fixed-template calibration, and suggests practical variants such as
 60 score filtering, cluster-level aggregation, and template-specific calibration (Section 5.4, Section 4.5,
 61 Appendix XIV). Across COCO, VOC, Open Images [7], LVIS [6], and four detectors (Ground-
 62 ingDINO, OWL-ViT, YOLO-World, RegionCLIP), in the regimes where the null-pool assumption
 63 is auditable the procedure controls per-family FDR; in regimes with non-exhaustive annotation or
 64 score mixing, the observed failures match the mechanisms analyzed by the rejection condition. A
 65 stratified concentration analysis yields a non-vacuous pooled FDP bound on the majority of families
 66 (Corollary 4).

67 2 Related Work

68 Self-normalized conformal e-value validity and e-BH FDR control are standard in the e-value liter-
 69 ature. Our contribution is to instantiate them for OVD hallucination filtering, to make the resulting
 70 e-BH rejection condition explicit for detection families (Theorem 1), and to pair it with a contam-
 71 ination analysis (Theorems 2–3) relevant to non-exhaustive annotation. The condition is algebraic
 72 but useful for diagnosing power loss in large, contaminated, or score-mixed families. The main
 73 distinction from prior conformal detection work is that we control released-box FDR for dependent
 74 candidate boxes within an image-query family, rather than coverage or recall-oriented risk.

75 **OVD reliability.** OVD detectors [9–22] and open-set/open-world variants [23–29] target new-
 76 class coverage rather than reliability of released detections. OVD-specific hallucinations and fine-
 77 grained failures are documented in [1, 30]; BIRDet [31] modifies detector representations, whereas
 78 we wrap any frozen detector with a post-hoc finite-sample guarantee.

79 **Conformal multiple testing and e-values.** Conformal prediction [32] and conformal risk con-
 80 trol [33] give distribution-free coverage or risk bounds. In vision, SeqCRC [4] controls sequential
 81 detection risk and is recall-/loss-oriented, while Conf-OT [5] adapts conformal prediction to zero-
 82 shot image-label classification; other applications target bounding-box uncertainty [34] and seg-
 83 mentation [35]. These methods have different targets than ours: none controls family-level FDR
 84 for dependent candidate boxes released by an OVD pipeline. Standard BH [2]/BY [3] applied to
 85 conformal p-values requires independence or PRDS and acts at the wrong granularity. We instead
 86 build on e-values [36], e-BH [8], and betting calibration [37], but our contribution is not their appli-
 87 cation: Theorem 1 establishes a closed-form activation frontier characterizing *when* e-BH can reject
 88 in an image-query family, and Corollary 1 shows self-normalized e-BH saturates it. These are not
 89 consequences of the e-BH guarantee, which bounds FDR but says nothing about power. Confor-
 90 mal e-values have appeared in novelty detection [38], but not OVD hallucination filtering. Other
 91 directions, including adaptive conformal methods [39], learned-score FDR [40], side-information
 92 thresholding [41], limited-FP prediction sets [42], and defer-to-expert systems [43], do not address
 93 family-level FDR under strong within-family dependence.

94 **Label contamination and missing annotation.** Missing-label detection work [44, 45] motivates
 95 our contamination model (Section 6) but addresses training, not post-hoc control. Contamination
 96 and group-shift results [46, 47] are complementary to our $1/\eta$ saturation ceiling and graceful-
 97 degradation analysis: they characterize how validity degrades under an impure null pool, while
 98 the rejection condition of Theorem 1 indicates when that degradation becomes a zero-power phase
 99 transition for OVD families.

100 3 Problem Setup

101 Let x denote an image and let q denote a text prompt, such as a category name or a free-form phrase.
 102 Fix a frozen open-vocabulary detector together with its score thresholding and non-maximum sup-
 103 pression pipeline, and treat the whole stack as a deterministic candidate generator

$$G_f(x, q) = \{(b_k, s_k)\}_{k=1}^{K(x, q)}, \quad (1)$$

104 where the subscript $f = (x, q)$ indexes the image-query *family*, b_k is a candidate box, $s_k \in \mathbb{R}$ is its
 105 detector score, and $K(x, q)$ is the data-dependent family size after post-processing.

106 We observe a calibration dataset

$$\mathcal{D}_{\text{cal}} = \{(x_i, q_i, B_i^*)\}_{i=1}^n, \quad (2)$$

107 where B_i^* is the set of ground-truth boxes matching prompt q_i in image x_i . The statistical assumption
 108 is that, under the null hypothesis for a given test candidate, its score is exchangeable with the null
 109 scores from the calibration split, conditional on the fit split. This is a within-null-pool assumption;
 110 it does *not* require exchangeability across families or between true positives and nulls. Within a
 111 family we allow arbitrary dependence among candidate boxes, which matters in this setting because
 112 proposal sharing and NMS induce strong and unmodelable dependence. Pooling null scores across
 113 families is justified when they share a common null score distribution; our LOO diagnostic (mean

114 = 1.000 on all configurations) is consistent with this assumption. Appendix III gives a formal
 115 approximate-exchangeability bound under mild groupwise shift.

116 For a candidate b_k from a test family (x, q) , define the null hypothesis

$$H_k^0 : b_k \text{ is unmatched under greedy IoU matching at threshold } \tau, \quad (3)$$

117 where greedy matching assigns each ground-truth box to at most one detection (highest IoU first; see
 118 Appendix XII for details). A true null thus corresponds to a hallucinated detection. We use $\tau = 0.5$
 119 throughout (the standard COCO criterion) and verify robustness to $\tau \in \{0.3, 0.7\}$ in Appendix VI.
 120 When $B^* = \emptyset$, every candidate in the family is null.

121 **Query-vocabulary annotation.** The null pool requires “absent in the annotation” to imply “truly
 122 absent” only for the *queried categories*. This is strictly weaker than full per-class exhaustive anno-
 123 tation and applies whenever the deployment vocabulary is known at calibration time.

124 The goal is a selection rule $\mathcal{S}$ that releases a subset of candidates while controlling

$$\text{FDR}(\mathcal{S}) = \mathbb{E} \left[\frac{|\{k : \mathcal{S}_k = 1, H_k^0 \text{ true}\}|}{|\{k : \mathcal{S}_k = 1\}| \vee 1} \right] \leq \alpha \quad (4)$$

125 for a user-specified level $\alpha \in (0, 1)$, in finite samples, without any within-family independence
 126 assumption.

127 Family-level FDR is the primary target because each image-query pair is a natural invocation of the
 128 OVD system; this also prevents a few large families from dominating the risk. Pooled FDP, the ra-
 129 tio of false discoveries to total releases across families, is reported as an empirical global-precision
 130 diagnostic, with a non-vacuous concentration certificate on the $K \leq 5$ stratum (Corollary 4). The ex-
 131 changeability assumption is a calibration assumption on the null pool, not an independence assump-
 132 tion among detections; it may fail under prompt- or dataset-level shifts, and our LOO diagnostic
 133 detects only gross violations.

134 4 Method

135 The procedure has three components: labeling TP and null scores on a fit split, calibrating self-
 136 normalized e-values on a null-only split, and applying e-BH within each image-query family on the
 137 test split. The detector is not modified and no within-family independence is assumed.

138 4.1 Three-Split Protocol

139 The procedure uses three disjoint family-level splits. On the *fit* split (60% of families), we apply
 140 greedy IoU matching (§3) to label candidates as TP ($y_{i,k} = 1$) or null, and learn a score-to-evidence
 141 map $\phi(s)$. On the *calibration* split (20%), we retain null candidates only; these are used solely to
 142 self-normalize test e-values and never to fit ϕ . On the *test* split (20%), we compute one e-value per
 143 candidate and apply e-BH within each family. Splits are taken by family identifier so intra-family
 144 dependence is preserved.

145 The fit split yields a nonnegative score transform $\phi : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$, interpretable as a density-ratio es-
 146 timate $\phi(s) \approx \hat{p}_1(s)/\hat{p}_0(s)$. Our default is an unsmoothed 80-bin histogram (KDE/logistic/isotonic
 147 ablations in Appendix V); validity requires only that ϕ be deterministic given the fit split, so estima-
 148 tor quality affects power, not validity.

149 4.2 Self-Normalized Conformal Betting e-Values

150 The self-normalized e-value below builds on conformal e-value constructions [8, 36]; the detection-
 151 specific power theory (Theorem 1) that follows is new.

152 Fix a test candidate score S whose null hypothesis is true, and let

$$S_1^{\text{cal}}, \dots, S_m^{\text{cal}} \quad (5)$$

153 be the null scores from the calibration split. Conditional on the fit split, assume the null scores are
 154 exchangeable:

$$(S, S_1^{\text{cal}}, \dots, S_m^{\text{cal}}) \text{ exchangeable under } H^0. \quad (6)$$

155 Define the self-normalized e-value

$$e(S) = \frac{(m+1)\phi(S)}{\phi(S) + \sum_{j=1}^m \phi(S_j^{\text{cal}})}. \quad (7)$$

156 Relative to mean-centering alternatives, this construction absorbs calibration randomness into the
 157 denominator itself and therefore yields an exact finite-sample validity statement rather than a high-
 158 probability approximation. It is also closely aligned with the betting/e-value viewpoint in modern
 159 sequential inference [36, 37].

160 **Lemma 1** (Exact validity of the self-normalized e-value). *Condition on the fit split and suppose*
 161 *$(S, S_1^{\text{cal}}, \dots, S_m^{\text{cal}})$ is exchangeable under H^0 . Let ϕ be any deterministic nonnegative function of*
 162 *the score. Then the e-value in (7) is valid:*

$$\mathbb{E}_{H^0}[e(S)] \leq 1. \quad (8)$$

163 *In fact, if $\phi(S) + \sum_{j=1}^m \phi(S_j^{\text{cal}}) > 0$ a.s., equality holds ($\mathbb{E}_{H^0}[e(S)] = 1$).*

164 *Proof sketch.* By exchangeability, $\phi(S_j)/T$ have equal expectation for $j = 0, \dots, m$ where $T =$
 165 $\sum_{j=0}^m \phi(S_j)$; summing gives $\mathbb{E}[\phi(S_0)/T] = 1/(m+1)$. Full proof in Appendix III. Validity is
 166 unconditional on ϕ ; the role of ϕ is purely to shape power, enabling the detection-specific power
 167 theory that follows.

168 4.3 Per-Family e-BH for Arbitrary Within-Family Dependence

169 Consider a test family with candidate e-values $e_1, \dots, e_K$. We apply e-BH at level α within that
 170 family. Let $e_{(1)} \geq \dots \geq e_{(K)}$ denote the sorted e-values and define

$$\hat{k} = \max \left\{ k \in \{1, \dots, K\} : e_{(k)} \geq \frac{K}{\alpha k} \right\}, \quad (9)$$

171 with the convention $\hat{k} = 0$ if the set is empty. The rejection set is then

$$\mathcal{R} = \{j : e_j \geq e_{(\hat{k})}\}. \quad (10)$$

172 We work with e-values rather than p-values to sidestep the dependence issue. Standard BH [2]
 173 requires independence or positive dependence assumptions [3] that are difficult to justify after pro-
 174 posal generation and NMS, whereas e-BH [8] accommodates arbitrary dependence once the input
 175 e-values are valid.

176 **Proposition 1** (Per-family FDR control). *Suppose every null candidate in a test family is assigned*
 177 *a valid e-value by (7). Then e-BH at level α satisfies*

$$\mathbb{E} \left[\frac{|\mathcal{R} \cap \mathcal{H}_0|}{|\mathcal{R}| \vee 1} \right] \leq \alpha \quad (11)$$

178 *for that family, regardless of the dependence structure among the candidates in the family.*

179 Averaging over the test-family distribution immediately gives a marginal family-averaged FDR guar-
 180 antee as well. Appendix III also gives a hierarchical extension: construct family-level e-values from
 181 an independent split, run e-BH across families, and then run within-family e-BH on the selected
 182 families. This controls family selection FDR together with selected-family mean FDP.

183 4.4 Activation Frontier for Image-Query Families

184 Lemma 1 and Proposition 1 establish validity. Power, i.e., whether e-BH actually rejects any candi-
 185 dates in a family, requires a separate characterization.

186 Theorem 1 below gives the necessary and sufficient condition in one line: *the r -th rejection in a*
 187 *family of size K is possible iff the r -th largest evidence exceeds $K C_m / [\alpha r(m+1) - K]$. Several*
 188 *consequences used below follow from this condition.*

Theorem 1 (Family activation frontier). *Consider a test family of size K with candidate ϕ -values sorted as $\phi_{(1)} \geq \phi_{(2)} \geq \dots \geq \phi_{(K)}$. Let $C_m = \sum_{j=1}^m \phi(S_j^{\text{cal}})$ denote the calibration pool aggregate. Then e-BH at level α rejects at least r candidates in this family if and only if*

$$\phi_{(r)} \geq T_r(C_m) := \frac{K C_m}{\alpha r (m+1) - K}, \quad (12)$$

provided $\alpha r (m+1) > K$. If $\alpha r (m+1) \leq K$, the r -th rejection is structurally impossible regardless of the evidence.

The proof follows by substituting the e-value definition into the e-BH criterion and rearranging; see Appendix III.

Corollary 1 (Frontier saturation). *Call a true positive frontier-reachable in family g if it is among the top- R_g candidates by ϕ , where $R_g = \max\{r : \phi_{(r,g)} \geq T_r(C_m)\}$ is the maximum number of rejections permitted by Theorem 1. Then self-normalized e-BH rejects a candidate if and only if it is frontier-reachable.*

The identity is algebraic. Since $e(S) = (m+1)\phi(S)/[\phi(S) + C_m]$ is strictly increasing in ϕ , sorting by e-value coincides with sorting by ϕ , and the e-BH selection count equals R_g (Appendix III). Consequently, the gap between observed recall and 100% reflects TPs whose density ratios fall below the family activation threshold, which is a property of the detector’s score geometry rather than of the procedure. Improving recall therefore requires either improving ϕ (sharper estimator or stronger detector scores) or reducing the effective family size K (Remark 1), not the testing procedure. Theorem 1 and Corollary 1 together explain binary e-values’ powerlessness, high- K families’ stricter thresholds, and contamination’s effect on C_m (Section 6).

Remark 1 (Frontier descent via deterministic pre-screening). Theorem 1 shows that T_r grows with family size K . A deterministic score floor s_{floor} , fixed a priori or chosen on the fit split, removes candidates with score $< s_{\text{floor}}$ from each test family. The result is an effective size $K_{\text{eff}} \leq K$ and strictly lower activation thresholds. Validity is preserved: each surviving candidate’s e-value (7) depends only on the calibration pool (unchanged) and its own ϕ , so Lemma 1 applies to any deterministic subset. The recall trade-off is transparent: the floor discards TPs with score below s_{floor} , but the surviving TPs face a lower activation barrier. The net effect is a substantial recall gain whenever the detector assigns sufficiently distinct scores to TPs and nulls (Section 5.4).

Implementation details. We split by family identifier (not by candidate) so that intra-family dependence is preserved. Only null candidates enter the calibration pool (7), keeping exchangeability exact. The detector is frozen; we learn only a one-dimensional score map (unsmoothed histogram by default).

4.5 Cluster-level e-values under NMS-induced dependence

Detectors without internal NMS produce 5–20 overlapping candidates per object, diluting per-candidate budget. We partition candidates by the undirected NMS graph ($\text{IoU} \geq \tau$, connected components) and assign one e-value per cluster C :

$$E_{\text{unif}}(C) = \frac{1}{|C|} \sum_{i \in C} E_i, \quad E_{\text{cal}}(C) = \frac{(n+1) R(C)}{R(C) + \sum_{j=1}^n R(C_j^{\text{cal}})}, \quad R(C) = \max_{i \in C} \phi(s_i), \quad (13)$$

where E_i is (7) and $\{C_j^{\text{cal}}\}$ is a null-only pool of calibration clusters exchangeable with test clusters under H_0 .

Proposition 2 (Cluster-level validity). $\mathbb{E}_{H_0}[E_{\text{unif}}(C)] \leq 1$ (closure under predictable convex combinations) and $\mathbb{E}_{H_0}[E_{\text{cal}}(C)] \leq 1$ (self-normalized construction of Vovk and Wang [36] at cluster granularity). Per-family e-BH on either e-value controls TP-cluster FDR at α (Appendix XXI).

Proposition 2 establishes cluster-level validity. Empirically, predictable uniform aggregation is conservative, whereas calibrated score-adaptive weighting improves object-level power: uniform averaging dilutes a cluster with one strong and ten weak members, while calibrating R_{max} against null clusters does not. When τ is so high that every cluster is a singleton, E_{cal} reduces exactly to (7) and gracefully recovers the candidate baseline.

α	Betting e-BH (ours)				Baselines	
	Rej.	Per-family FDP	Pooled FDP	Recall	Pooled FDP (p-BH)	Recall (binary)
0.05	459.6	0.43%	4.9%	12.5%	17.9%	0.16%
0.10	615.1	0.78%	6.4%	16.4%	27.1%	0.18%
0.15	731.4	1.10%	8.6%	19.1%	35.0%	0.19%
0.20	823.0	1.45%	10.1%	21.1%	42.3%	0.83%

Table 1: Full-scale COCO val (4,952 images) with GroundingDINO, 20 random splits. Per-family FDP (Proposition 1) $\ll \alpha$ at every level; pooled FDP is diagnostic at full-family scale; a stratified empirical Bernstein bound (Corollary 4) is non-vacuous on the $K \leq 5$ stratum (57% of families). p-value BH violates pooled target; binary e-BH has negligible recall. Pass rate at $\alpha = 0.10$: 19/20.

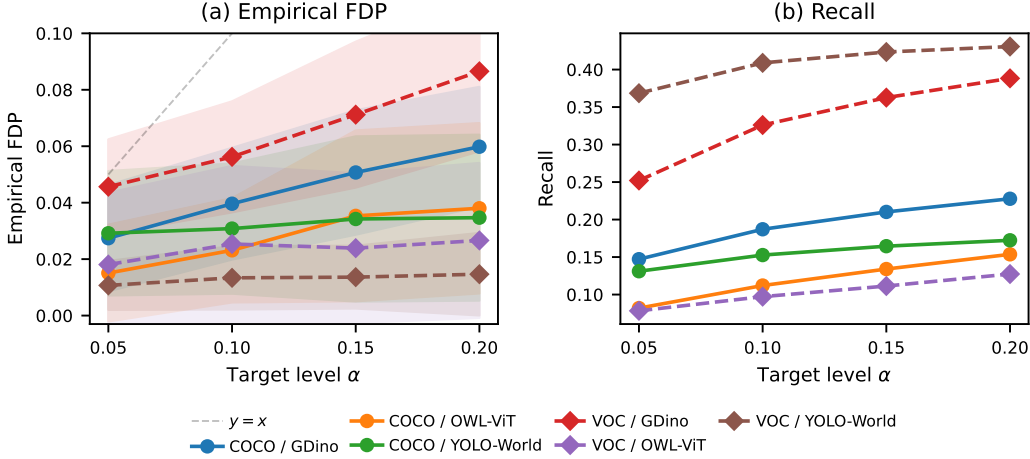


Figure 2: Risk curves ($\alpha \in \{0.05, \dots, 0.20\}$, six configs, 1,000-image subsets, 20 splits; full-scale in Table 1). (a) FDP stays below $y = x$. (b) Recall grows with α . Solid: COCO; dashed: VOC.

5 Experiments

We evaluate whether the proposed wrapper controls per-family FDR on frozen OVD detectors while retaining non-trivial recall.

5.1 Setup

We evaluate on COCO val 2017 [48] (4,952 images) and Pascal VOC 2012 val [49], both exhaustively annotated. Three detectors span distinct architectures: GroundingDINO [50], OWL-ViT [51], and YOLO-World [52]. Each image is paired with one present and two absent prompts (3,000 families per 1k-image configuration), split 60/20/20 (fit/cal/test) by family. The score map ϕ is an unsmoothed 80-bin histogram. Numbers are means $\pm$ s.d. over 20 random splits unless noted. Appendix XII specifies the prompt set, template bank, detector score thresholds and NMS settings, random seeds (0–19), split construction, and histogram binning used in all experiments.

5.2 Main Result: Per-Family FDR Control on COCO and VOC

Table 1 reports the core evaluation on full COCO val with GroundingDINO. The primary metric is per-family FDR (Proposition 1); pooled FDP is diagnostic. At $\alpha = 0.10$, per-family mean FDP is well below nominal across all six configurations (Table 9; max 1.69%), pooled FDP is 6.44% (19/20 pass), recall is 16.4%, and released detections have 93.6% empirical pooled precision vs. 72.9% for p -value BH; LOO mean is 1.000. Cross-detector 1k-subset coverage in App. VIII.

Figure 2 shows full risk curves across $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$: empirical FDP stays strictly below the nominal level and recall grows monotonically with α . OWL-ViT’s lower FDP reflects its smaller families ($K \approx 2.5$ vs. 10).

Method	Rej.	Pooled FDP	Per-family FDP	Recall	Pass?
Raw detector (no filter)	6164	88.7%	—	100% [†]	×
Cal. threshold [33, 42]	240	10.0%	7.09%	31.6%	×
p-value BH [2, 46]	383	27.7%	5.54%	—*	×
p-value BY [3, 46]	212	11.3%	3.29%	27.7%	△
Top- k per family (cal-tuned) [E2]	537	67.4%	67.4%	25.5%	×
CRC (mean-family risk) [53]	521	33.9%	9.73%	49.1%	×
Storey-BH [54]	514	39.7%	20.3%	41.1%	××
BKY two-stage [55]	414	30.2%	18.2%	39.0%	××
IHW-lite (cov. K) [56]	209	8.07%	7.90%	27.5%	△
Binary e-BH [8, 36]	2	1.6%	0.08%	0.2%	✓
Betting e-BH (ours)	134	3.96%	0.65%	18.7%	✓

Table 2: Baseline comparison, COCO-1k/GDino, $\alpha = 0.10$, 20 seeds. ✓/△/×/×× = all pass / partial / pooled fails / pooled and per-family both fail. *pooled FDP violated; [†]candidate-level recall is trivially 100% when all candidates are released. [E2]: top- k baseline picks largest k with cal mean family-FDP $\leq \alpha$ (Appendix XXII). IHW-lite uses family size K (binned) as the covariate; see Appendix XXII for the full per-config breakdown.

Config	s_{floor}	Rej.	Per-fam. FDP	Pooled FDP	Recall	Pass	Gain
COCO/GDino	0.05 (adaptive)	134	0.65%	4.0%	18.7%	1.00	—
	0.40 (fixed)	261	2.59%	11.5%	33.3%	0.30	1.78×
VOC/GDino	0.05 (adaptive)	196	1.69%	5.6%	32.6%	1.00	—
	0.40 (fixed)	317	4.42%	12.0%	49.0%	0.40	1.50×

Table 3: Score-floor pre-screening at $\alpha = 0.10$ (1k images, GDino, 20 seeds). Per-family FDP $\ll \alpha$ at every floor (Proposition 1 holds finite-sample). Pooled FDP and pass rate (fraction of seeds with pooled FDP $\leq \alpha$) reveal the deployment trade: the fit-split-selected *adaptive* floor (≈ 0.05 , chosen on fit nulls only) is the principled default; the aggressive fixed 0.40 trades 1.5–1.8× recall for pooled-FDP slack at $\alpha = 0.10$.

On the 1,000-image subsets (Figure 2), all six configurations attain 20/20 pooled-FDP pass at $\alpha \geq 0.10$ (Appendix IV); the full-scale COCO/GDino row in Table 1 is 19/20 at $\alpha = 0.10$. COCO-train confirms with exact LOO validity (Appendix XI). By Corollary 1, the residual recall gap is a detector-side limitation; score-floor pre-screening lifts recall to 33.3%/49.0% on COCO/VOC (Section 5.4). Pooled FDP admits a finite-sample concentration bound (Proposition 5); an empirical Bernstein refinement on the $K \leq 5$ stratum yields a non-vacuous certificate $\varepsilon = 0.074$ (Appendix IX).

5.3 Comparison with Thresholding and p -Value Procedures

We compare against a broad set of thresholding, p -value, risk-control, and adaptive multiple-testing baselines using the same candidate pools and calibration splits. Table 2 summarizes the comparison.

For p -value baselines, we compute conformal p -values $p(s) = (1 + |\{j : S_j^{\text{cal}} \geq s\}|)/(m+1)$ and apply BH/BY [2, 3] per family. *Calibration-tuned thresholding* picks the lowest threshold with calibration FDP $\leq \alpha$; it reaches 31.6% recall but fails 13/20 splits, since the cal-to-test concentration correction ($\varepsilon \approx 0.12$) consumes the budget. p -BH and BY violate the pooled target (27.7%/11.3%); binary e-BH is valid but powerless. VOC/GDino shows the same pattern; results are robust to estimator and cal pool size (App. V, VII).

5.4 Score-Floor Pre-Screening: Activation-Frontier Descent

A deterministic pre-screen shrinks K_{eff} and hence T_r (Remark 1). The fit-split-selected *adaptive* floor ($s_{\text{floor}}^* \approx 0.05$ –0.10, no test labels) is the pooled-safe default; the fixed 0.40 floor is a recall-oriented operating point that consumes pooled-FDP slack (Table 3). Per-family FDR holds at both; the choice maps to deployment Mode A vs B/C (App. XX).

Object-level recall via cluster e-BH. The calibrated cluster e-value E_{cal} (Section 4.5, $\tau = 0.5$, R_{max}) lifts recall 1.4–1.8× at matched FDP: at $\alpha = 0.10$, 18.7% $\rightarrow$ 32.1% COCO/GDino, 32.6% $\rightarrow$

54.3% VOC/GDino, 3.7% $\rightarrow$ 21.6% COCO/RegionCLIP ($5.8\times$, no internal NMS; App. XXVII). Stratified diagnostics by score quantile, object size, and category (App. XXVI) show no systematic calibration failure: rejections concentrate in the top score quintile, object-size pooled FDP is at or near α on COCO/GDino across sizes, and per-family FDR holds within every category; per-category pooled FDP is a stress diagnostic only.

6 A Three-Regime Deployment Map

The rejection condition (Theorem 1) and contamination saturation (Theorem 2) define three regimes with distinct failure modes (Figure 1, middle panel).

6.1 Regime I — Exhaustive Annotation (Feasible)

On COCO/VOC, α -validity holds and power is governed by T_r (baseline recall 16–41%, Table 1; Cor. 1); frontier-descent fixes lift recall to 33%/49% (score floor) and 1.4 – $5.8\times$ (cluster).

6.2 Regime II — Bounded Contamination (Boundary)

Non-exhaustive annotation contaminates the null pool with true positives.

Theorem 2 (Contamination saturation of the density ratio). *Let $r(s) = p_1(s)/p_0(s)$ be the clean density ratio. If an η fraction of the observed null pool is unlabeled positives, so $q_0 = (1-\eta)p_0 + \eta p_1$, then*

$$\phi_\eta(s) = \frac{p_1(s)}{q_0(s)} = \frac{r(s)}{1 - \eta + \eta r(s)} \quad (14)$$

satisfies $\phi_\eta \leq 1/\eta$ and compresses high-evidence regions; η is identified from $r_{\max}$, $M = \sup \phi_\eta$ via $\eta = (r_{\max} - M)/[M(r_{\max} - 1)]$.

Theorem 3 (True-label FDR under contamination). *For $q_0 = (1 - \eta)p_0 + \eta p_1$, any q_0 -valid e-value satisfies $\mathbb{E}_{p_0}[E] \leq 1/(1 - \eta)$; e-BH at level α thus controls true-label FDR at $\alpha/(1 - \eta)$, restored to α under level $\alpha(1 - \bar{\eta})$ when $\eta \leq \bar{\eta}$, with a zero-rejection phase transition when $1/\eta < T_1(C_m)$ (Cor. 3).*

Controlled contamination. Hiding $\eta_{\text{fam}} \in \{0, \dots, 0.50\}$ on COCO/VOC validates Theorem 2 (Fig. 4, Table 14): $\phi_{\max}$ tracks $1/\eta$, true FDP stays controlled, recall collapses. Three LVIS repairs fail (App. I); $\hat{\eta}$ -correction (38 rejections, 0.99% FDP) and vocabulary curation (App. II, 11–21% recall) address opposite ends of the chain.

6.3 Regime III — Fixed-Template Deployment (Score-Mixing Compression)

Fixed template banks create a different bottleneck: *score-mixing compression* (Proposition 3) drives $\sup \phi_{\text{pool}} \approx 14 \ll T_1 \approx 283$ under pooling; per-template fitting recovers 98.6 rejections at 3.9% FDP (App. XIV). Template count is bounded by calibration ($T_{\text{crit}} \propto N_{\text{img}}/K_{\text{min}}^{\text{cal}}$).

7 Discussion and Conclusion

We presented a post-hoc reliability wrapper for frozen open-vocabulary detectors that controls family-level hallucination FDR for auditable category-name or fixed-template prompts. The statistical mechanism is simple: self-normalized conformal betting e-values give finite-sample validity under null-score exchangeability, and e-BH gives dependence-robust FDR control within each image-query family. The diagnostic contribution is the activation frontier $T_r(K, C_m)$, which separates validity from power and explains three otherwise confusing phenomena: binary e-values can be valid but powerless, large candidate families impose high evidence thresholds, and contaminated null pools create a zero-power phase transition. This view yields three practical variants—adaptive score floors, cluster aggregation, and template-specific calibration—each addressing one bottleneck. Deployment distinguishes three modes: per-family FDR control, empirical pooled-FDP/LOO diagnostics, and Bernstein-certified pooled-FDP bounds on the $K \leq 5$ stratum. The method is a release filter, not a detector improvement; validity depends on auditable absence labels and deployment-specific contamination bounds.

References

- [1] Frank Ruis, Gertjan J. Burghouts, and Hugo Kuijf. Fantastic tractor-dogs and how not to find them with open-vocabulary detectors. In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=jUuXNrG7wh>.
- [2] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal statistical society: series B (Methodological)*, 57(1):289–300, 1995.
- [3] Yoav Benjamini and Daniel Yekutieli. The control of the false discovery rate in multiple testing under dependency. *Annals of statistics*, pages 1165–1188, 2001.
- [4] Léo Andéol, Luca Mossina, Adrien Mazoyer, and Sébastien Gerchinovitz. Conformal Object Detection by Sequential Risk Control. working paper or preprint, December 2025. URL <https://hal.science/hal-05399935>.
- [5] Julio Silva-Rodríguez, Ismail Ben Ayed, and Jose Dolz. Conformal prediction for zero-shot models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 19931–19941, 2025.
- [6] Agrim Gupta, Piotr Dollar, and Ross Girshick. Lvis: A dataset for large vocabulary instance segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5356–5364, 2019.
- [7] Alina Kuznetsova, Hassan Rom, Neil Alldrin, Jasper Uijlings, Ivan Krasin, Jordi Pont-Tuset, Shahab Kamali, Stefan Popov, Matteo Mallocci, Alexander Kolesnikov, et al. The open images dataset v4: Unified image classification, object detection, and visual relationship detection at scale. *International journal of computer vision*, 128(7):1956–1981, 2020.
- [8] Ruodu Wang and Aaditya Ramdas. False discovery rate control with e-values. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 84(3):822–852, 2022.
- [9] Xiuye Gu, Tsung-Yi Lin, Weicheng Kuo, and Yin Cui. Open-vocabulary object detection via vision and language knowledge distillation. *arXiv preprint arXiv:2104.13921*, 2021.
- [10] Yiwu Zhong, Jianwei Yang, Pengchuan Zhang, Chunyuan Li, Noel Codella, Liunian Harold Li, Luowei Zhou, Xiyang Dai, Lu Yuan, Yin Li, et al. Regionclip: Region-based language-image pretraining. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 16793–16803, 2022.
- [11] Liunian Harold Li, Pengchuan Zhang, Haotian Zhang, Jianwei Yang, Chunyuan Li, Yiwu Zhong, Lijuan Wang, Lu Yuan, Lei Zhang, Jenq-Neng Hwang, et al. Grounded language-image pre-training. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10965–10975, 2022.
- [12] Xingyi Zhou, Rohit Girdhar, Armand Joulin, Philipp Krähenbühl, and Ishan Misra. Detecting twenty-thousand classes using image-level supervision. In *European conference on computer vision*, pages 350–368. Springer, 2022.
- [13] Yu Du, Fangyun Wei, Zihe Zhang, Miaoqing Shi, Yue Gao, and Guoqi Li. Learning to prompt for open-vocabulary object detection with vision-language model. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 14084–14093, 2022.
- [14] Weicheng Kuo, Yin Cui, Xiuye Gu, AJ Piergiovanni, and Anelia Angelova. F-vlm: Open-vocabulary object detection upon frozen vision and language models. *arXiv preprint arXiv:2209.15639*, 2022.
- [15] Xiaoshi Wu, Feng Zhu, Rui Zhao, and Hongsheng Li. Cora: Adapting clip for open-vocabulary detection with region prompting and anchor pre-matching. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 7031–7040, 2023.

- [16] Lewei Yao, Jianhua Han, Xiaodan Liang, Dan Xu, Wei Zhang, Zhenguo Li, and Hang Xu. Detclipv2: Scalable open-vocabulary object detection pre-training via word-region alignment. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23497–23506, 2023.
- [17] Hao Zhang, Feng Li, Xueyan Zou, Shilong Liu, Chunyuan Li, Jianwei Yang, and Lei Zhang. A simple framework for open-vocabulary segmentation and detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 1020–1031, 2023.
- [18] Jiong Wang, Huiming Zhang, Haiwen Hong, Xuan Jin, Yuan He, Hui Xue, and Zhou Zhao. Open-vocabulary object detection with an open corpus. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 6759–6769, 2023.
- [19] Jiaming Li, Jiacheng Zhang, Jichang Li, Ge Li, Si Liu, Liang Lin, and Guanbin Li. Learning background prompts to discover implicit knowledge for open vocabulary object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 16678–16687, 2024.
- [20] Shiyu Zhao, Samuel Schuster, Long Zhao, Zhixing Zhang, Yumin Suh, Manmohan Chandraker, Dimitris N Metaxas, et al. Taming self-training for open-vocabulary object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13938–13947, 2024.
- [21] Dahun Kim, Anelia Angelova, and Weicheng Kuo. Region-centric image-language pretraining for open-vocabulary detection. In *European Conference on Computer Vision*, pages 162–179. Springer, 2024.
- [22] Jang Hyun Cho and Philipp Krähenbühl. Language-conditioned detection transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16593–16603, 2024.
- [23] KJ Joseph, Salman Khan, Fahad Shahbaz Khan, and Vineeth N Balasubramanian. Towards open world object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5830–5840, 2021.
- [24] Jiaming Han, Yuqiang Ren, Jian Ding, Xingjia Pan, Ke Yan, and Gui-Song Xia. Expanding low-density latent regions for open-set object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9591–9600, 2022.
- [25] Hiran Sarkar, Vishal Chudasama, Naoyuki Onoe, Pankaj Wasnik, and Vineeth N Balasubramanian. Open-set object detection by aligning known class representations. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 219–228, 2024.
- [26] Heng Yin, Yuqiang Ren, Ke Yan, Shouhong Ding, and Yongtao Hao. Rod-mlm: Towards more reliable object detection in multimodal large language models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 14358–14368, 2025.
- [27] Xing Xi, Yangyang Huang, Ronghua Luo, and Yu Qiu. Ow-ovd: Unified open world and open vocabulary object detection. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 25454–25464, 2025.
- [28] Shan Zhang, Yao Ni, Jinhao Du, Yuan Xue, Philip Torr, Piotr Koniusz, and Anton Van Den Hengel. Open-world objectness modeling unifies novel object detection. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 30332–30342, 2025.
- [29] Muli Yang, Gabriel James Goenawan, Huaiyuan Qin, Kai Han, Xi Peng, Yanhua Yang, and Hongyuan Zhu. Detecting open world objects via partial attribute assignment. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 20318–20328, 2025.
- [30] Lorenzo Bianchi, Fabio Carrara, Nicola Messina, Claudio Gennaro, and Fabrizio Falchi. The devil is in the fine-grained details: Evaluating open-vocabulary object detectors for fine-grained understanding. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 22520–22529, 2024.

- [31] Ruizhe Zeng, Lu Zhang, Xu Yang, and Zhiyong Liu. Boosting open-vocabulary object detection by handling background samples. In *International Conference on Neural Information Processing*, pages 274–289. Springer, 2024.
- [32] Anastasios N Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.
- [33] Anastasios N Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. *arXiv preprint arXiv:2208.02814*, 2022.
- [34] Alexander Timans, Christoph-Nikolas Straehle, Kaspar Sakmann, and Eric Nalisnick. Adaptive bounding box uncertainties via two-step conformal prediction. In *European Conference on Computer Vision*, pages 363–398. Springer, 2024.
- [35] Luca Mossina and Corentin Friedrich. Controlling false positives in image segmentation via conformal prediction. *arXiv preprint arXiv:2511.15406*, 2025.
- [36] Vladimir Vovk and Ruodu Wang. E-values: Calibration, combination and applications. *The Annals of Statistics*, 49(3):1736–1754, 2021.
- [37] Ian Waudby-Smith and Aaditya Ramdas. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 86(1):1–27, 02 2024. ISSN 1369-7412. doi: 10.1093/jrssb/qkad009. URL <https://doi.org/10.1093/jrssb/qkad009>.
- [38] Meshi Bashari, Amir Epstein, Yaniv Romano, and Matteo Sesia. Derandomized novelty detection with fdr control via conformal e-values. *Advances in Neural Information Processing Systems*, 36:65585–65596, 2023.
- [39] Isaac Gibbs, John J Cherian, and Emmanuel J Candès. Conformal prediction with conditional guarantees. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 87(4):1100–1126, 2025.
- [40] Ariane Marandon, Lihua Lei, David Mary, and Etienne Roquain. Machine learning meets false discovery rate. *arXiv preprint arXiv:2208.06685*, 2, 2022.
- [41] Lihua Lei and William Fithian. Adapt: an interactive procedure for multiple testing with side information. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 80(4):649–679, 2018.
- [42] Adam Fisch, Tal Schuster, Tommi Jaakkola, and Regina Barzilay. Conformal prediction sets with limited false positives. In *International Conference on Machine Learning*, pages 6514–6532. PMLR, 2022.
- [43] Yannis Montreuil, Shu Heng Yeo, Axel Carlier, Lai Xing Ng, and Wei Tsang Ooi. A two-stage learning-to-defer approach for multi-task learning. *arXiv preprint arXiv:2410.15729*, 2024.
- [44] Mengmeng Xu, Yancheng Bai, Bernard Ghanem, Boxiao Liu, Yan Gao, Nan Guo, Xiaochun Ye, Fang Wan, H You, D Fan, et al. Missing labels in object detection. In *CVPR workshops*, volume 3, 2019.
- [45] Yuewei Yang, Kevin J Liang, and Lawrence Carin. Object detection as a positive-unlabeled problem. *arXiv preprint arXiv:2002.04672*, 2020.
- [46] Stephen Bates, Emmanuel Candès, Lihua Lei, Yaniv Romano, and Matteo Sesia. Testing for outliers with conformal p-values. *The Annals of Statistics*, 51(1):149–178, 2023.
- [47] Rina Foygel Barber, Emmanuel J Candès, Aaditya Ramdas, and Ryan J Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845, 2023.
- [48] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *European conference on computer vision*, pages 740–755. Springer, 2014.

- 462 [49] Mark Everingham, Luc Van Gool, Christopher KI Williams, John Winn, and Andrew Zisser-
 463 man. The pascal visual object classes (voc) challenge. *International journal of computer*
 464 *vision*, 88(2):303–338, 2010.
- 465 [50] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chun-
 466 yuan Li, Jianwei Yang, Hang Su, et al. Grounding dino: Marrying dino with grounded pre-
 467 training for open-set object detection. In *European conference on computer vision*, pages
 468 38–55. Springer, 2024.
- 469 [51] Matthias Minderer, Alexey Gritsenko, Austin Stone, Maxim Neumann, Dirk Weissenborn,
 470 Alexey Dosovitskiy, Aravindh Mahendran, Anurag Arnab, Mostafa Dehghani, Zhuoran Shen,
 471 et al. Simple open-vocabulary object detection. In *European conference on computer vision*,
 472 pages 728–755. Springer, 2022.
- 473 [52] Tianheng Cheng, Lin Song, Yixiao Ge, Wenyu Liu, Xinggang Wang, and Ying Shan. Yolo-
 474 world: Real-time open-vocabulary object detection. In *Proceedings of the IEEE/CVF confer-*
 475 *ence on computer vision and pattern recognition*, pages 16901–16911, 2024.
- 476 [53] Anastasios Nikolas Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster.
 477 Conformal risk control. In *The Twelfth International Conference on Learning Representations*,
 478 2024. URL <https://openreview.net/forum?id=33XGfHLtZg>.
- 479 [54] John D Storey. A direct approach to false discovery rates. *Journal of the Royal Statistical*
 480 *Society Series B: Statistical Methodology*, 64(3):479–498, 2002.
- 481 [55] Yoav Benjamini, Abba M Krieger, and Daniel Yekutieli. Adaptive linear step-up procedures
 482 that control the false discovery rate. *biometrika*, pages 491–507, 2006.
- 483 [56] Nikolaos Ignatiadis, Bernd Klaus, Judith B Zaugg, and Wolfgang Huber. Data-driven hypoth-
 484 esis weighting increases detection power in genome-scale multiple testing. *Nature methods*,
 485 13(7):577–580, 2016.

I Repair Attempts on LVIS

We tested three targeted repairs on LVIS: (i) Mondrian splitting by absent/present status to isolate cleaner null strata; (ii) a minimum family-size filter ($k_{\min}$) to remove high-variance small families; (iii) a null-floor regularizer on ϕ to cap density-ratio blowup. None of the three restored FDR control. With $\phi_{\max} \approx 30$, the signal content of the null pool is already too degraded for e-BH to distinguish hallucinations from genuine detections at a controlled rate, and none of the three patches closes the annotation gap.

These results suggest that the limiting factor on LVIS is the quality of the null pool rather than the particular post-hoc tuning choice. They are consistent with the contamination mechanism analyzed in Theorem 2 and indicate that the oracle-null assumption is not satisfied on LVIS at the scale we evaluated.

II Honest Restricted Evaluation on LVIS and OpenImages

In-place statistical repairs (Appendix I) cannot restore validity on LVIS/OI because the annotation signal is the missing component. A complementary strategy is to *curate* the evaluation pool so that the oracle-null assumption holds: remove families whose absence cannot be certified, then run the unmodified pipeline. The contamination boundary is addressed from two sides. In-place $\hat{\eta} = 1/\phi_{\max}$ correction (Theorem 3, Section 6) adjusts the procedure for contamination that is estimated from the data, and vocabulary curation removes families for which the oracle-null assumption cannot be certified in the first place.

LVIS protocol. LVIS v1.0’s `exhaustive_category_ids` per-image annotation identifies categories that were exhaustively labeled. We retain a family (image, prompt) as a valid null if either (i) the image is annotated NEG for the prompt’s category, or (ii) the category appears in the image’s `exhaustive_category_ids`. This yields 536.6 families per split on average (20 seeds, 1000-image draws). Present families are unchanged, so this is strictly a subtraction on the absent side.

OpenImages V7 protocol. OI V7 ships image-level labels with explicit positive and negative verification flags. We admit a family only if the target category has a verified negative label on the image (image-label clean). This gives a pool of $\sim 1.0K$ families per split, predominantly present.

Results ($\alpha = 0.10$, 20 seeds). On both pools all four detectors pass α validity pooled and per-family:

Dataset	Detector	Rej.	Pooled FDP	Per-family FDP	Recall	Pool pass
LVIS (aux clean)	GDino	66.6	2.2%	0.4%	11.4%	20/20
	OWLv2	105.6	3.0%	1.2%	20.9%	20/20
	YOLO-World	31.2	3.7%	0.8%	12.6%	20/20
OI V7 (img-label clean)	GDino	54.5	2.1%	0.7%	12.3%	20/20

Interpretation. Pooled FDP drops from LVIS-full’s 17.7% (the baseline row of Table 19) to 2.2–3.7% under aux-clean restriction, at non-trivial recall (11–21%). OpenImages shows the same pattern at 2.1% FDP with 12.3% recall. These numbers are *not* comparable to the operational LVIS row in the main boundary table: the evaluation pools differ (aux-clean is a strict subset whose null certification is defensible; the full LVIS pool contains families whose absence is an assumption). What this appendix establishes is that the per-family validity the theory guarantees *is* recoverable on LVIS/OI, at the cost of evaluating a smaller, auditable slice of the benchmark. In-place $\hat{\eta}$ correction (which yields 38 rejections at 0.99% FDP on full LVIS; Section 6) and vocabulary curation are therefore complementary rather than substitutable.

III Proofs

Proof of Theorem 1. The e-value $e(S) = (m+1)\phi(S)/[\phi(S) + C_m]$ is strictly increasing in $\phi(S)$ for $C_m > 0$. The e-BH criterion for the r -th rejection requires $e_{(r)} \geq K/(\alpha r)$. Substituting

528 the e-value definition:

$$\frac{(m+1)\phi_{(r)}}{\phi_{(r)} + C_m} \geq \frac{K}{\alpha r}.$$

529 Rearranging gives $\phi_{(r)} [\alpha r (m+1) - K] \geq K C_m$. When $\alpha r (m+1) > K$, dividing yields (12).
 530 When $\alpha r (m+1) \leq K$, the e-BH threshold $K/(\alpha r) \geq m+1$ exceeds the supremum of $e(S)$,
 531 which is bounded by $m+1$. $\square$

532 **Proof of Corollary 1 (frontier saturation).** Since $e(S) = (m+1)\phi(S)/[\phi(S) + C_m]$ is strictly
 533 increasing in $\phi(S)$, sorting by e-value coincides with sorting by ϕ . The e-BH selection count in
 534 family g is $\hat{k}_g = \max\{k : \phi_{(k,g)} \geq T_k\} = R_g$ by Theorem 1, so self-normalized e-BH selects
 535 exactly the top- R_g candidates—every frontier-reachable candidate and no others. $\square$

536 **Proposition 3** (Score-mixing compression). *Under global calibration mixing T templates with null*
 537 *proportions $\pi_1, \dots, \pi_T$, the pooled density ratio decomposes as $\phi_{\text{pool}}(s) = \sum_t \mu_t(s) \phi_t(s)$ with*
 538 *null-posterior weights $\mu_t(s) = \pi_t q_{0,t}(s)/q_{0,\text{pool}}(s)$, assuming equal null and overall template pro-*
 539 *portions. Since $\mu_t(s) \rightarrow 0$ where $q_{0,t}(s)$ is small—typically where $\phi_t(s)$ peaks—sup ϕ_{pool} can fall*
 540 *below T_1 (Theorem 1) even when individual sup $\phi_t \gg T_1$.*

541 **Proof of Proposition 3.** Write $p_t, q_{0,t}$ for the overall and null score densities of template t . Under
 542 the equal-mixing assumption ($\omega_t = \pi_t$), $p_{\text{pool}} = \sum_t \pi_t p_t$ and $q_{0,\text{pool}} = \sum_t \pi_t q_{0,t}$. Substituting
 543 $p_t = \phi_t q_{0,t}$:

$$\phi_{\text{pool}}(s) = \frac{\sum_t \pi_t \phi_t(s) q_{0,t}(s)}{\sum_t \pi_t q_{0,t}(s)} = \sum_t \mu_t(s) \phi_t(s),$$

544 with $\mu_t(s) = \pi_t q_{0,t}(s)/q_{0,\text{pool}}(s) \geq 0$ and $\sum_t \mu_t(s) = 1$. Since $\phi_t(s) = p_t(s)/q_{0,t}(s)$ is large
 545 precisely when $q_{0,t}(s)$ is small, the corresponding $\mu_t(s)$ is small, so templates with high peak ϕ_t
 546 are down-weighted in the mixture. As $\phi_{\text{pool}}(s)$ is a convex combination, $\sup \phi_{\text{pool}} \leq \max_t \sup \phi_t$,
 547 with equality only when all templates share identical density ratios. $\square$

548 **Proof of Theorem 2.** (i) Since $r(s) \geq 0$, the denominator in (14) satisfies $1 - \eta + \eta r(s) \geq \eta r(s)$,
 549 giving $\phi_\eta(s) \leq 1/\eta$. (ii) Direct differentiation. (iii) Solve $M = r_{\max}/[1 - \eta + \eta r_{\max}]$ for η . $\square$

550 **Proof of Theorem 3.** Since $q_0 = (1 - \eta)p_0 + \eta p_1 \geq (1 - \eta)p_0$ pointwise, we have $p_0(s) \leq$
 551 $q_0(s)/(1 - \eta)$ for all s . Therefore

$$\mathbb{E}_{p_0}[E] = \int E(s) p_0(s) ds \leq \frac{1}{1 - \eta} \int E(s) q_0(s) ds \leq \frac{1}{1 - \eta}.$$

552 Thus $(1 - \eta)E$ is a valid e-value for the true null p_0 . Applying e-BH at level α to the original e-values
 553 is equivalent to applying e-BH at level $\alpha/(1 - \eta)$ to valid true-null e-values, yielding true-label FDR
 554 control at $\alpha/(1 - \eta)$. If $\eta \leq \bar{\eta}$ is known, then $(1 - \bar{\eta})E \leq (1 - \eta)E$ remains valid under p_0 , so
 555 e-BH at level $\alpha(1 - \bar{\eta})$ restores true-label FDR control at α . $\square$

556 **Lemma 2** (Approximate exchangeability yields graceful degradation [47]). *Condition on the fit*
 557 *split. Let P_{pool} be the pooled null score law and P_g the null score law for prompt/group g . Suppose*
 558 *the calibration scores are drawn from P_{pool} , while a null test score from group g is drawn from P_g ,*
 559 *and assume $\text{TV}(P_g, P_{\text{pool}}) \leq \delta$. Then the pooled self-normalized e-value satisfies*

$$\mathbb{E}_{P_g}[e(S)] \leq 1 + 2(m+1)\delta.$$

560 *Proof.* Let $F(S, S_1^{\text{cal}}, \dots, S_m^{\text{cal}})$ denote the self-normalized e-value. Under pooled exchangeability
 561 the joint law is $P_{\text{pool}}^{\otimes(m+1)}$ and Lemma 1 gives $\mathbb{E}[F] \leq 1$. Under group-specific shift the joint law is
 562 $P_g \otimes P_{\text{pool}}^{\otimes m}$. Since $0 \leq F \leq m+1$,

$$\left| \mathbb{E}_{P_g \otimes P_{\text{pool}}^{\otimes m}}[F] - \mathbb{E}_{P_{\text{pool}}^{\otimes(m+1)}}[F] \right| \leq 2(m+1) \text{TV}(P_g \otimes P_{\text{pool}}^{\otimes m}, P_{\text{pool}}^{\otimes(m+1)}) \leq 2(m+1)\delta.$$

563 The final inequality uses the tensorization $\text{TV}(P_g \otimes R, P_{\text{pool}} \otimes R) = \text{TV}(P_g, P_{\text{pool}})$ for a common
 564 factor R . Hence $\mathbb{E}_{P_g}[e(S)] \leq 1 + 2(m+1)\delta$. $\square$

565 **Corollary 2** (Per-family FDR under approximate exchangeability). *If every null in family g satisfies*
 566 *the bound in Lemma 2, then e-BH at level α obeys*

$$\text{FDR}_g \leq \alpha (1 + 2(m + 1)\delta).$$

567 *Equivalently, dividing the e-values by $1 + 2(m + 1)\delta$ restores exact per-family FDR control at level*
 568 *α .*

569 *Proof.* Set $\beta = 1 + 2(m + 1)\delta$ and define $\tilde{e} = e/\beta$. Then every null $\tilde{e}$ -value is valid by Lemma 2.
 570 Running e-BH at level α on the original e-values is equivalent to running e-BH at level $\alpha\beta$ on $\tilde{e}$, so
 571 the Wang-Ramdas guarantee yields $\text{FDR}_g \leq \alpha\beta$. $\square$ $\square$

572 **Corollary 3** (Power-collapse phase transition). *Under the contamination model of Theorem 2, if*

$$\frac{1}{\eta} < T_1(C_m) = \frac{K C_m}{\alpha (m + 1) - K},$$

573 *then $\sup_s \phi_\eta(s) < T_1(C_m)$, and e-BH produces zero rejections in the family. Contamination lowers*
 574 *the evidence ceiling $1/\eta$ while raising the activation threshold through C_m .*

575 *Proof.* Theorem 2 gives $\sup_s \phi_\eta(s) \leq 1/\eta$, and Theorem 1 shows that one rejection requires $\phi_{(1)} \geq$
 576 $T_1(C_m)$. If $1/\eta < T_1(C_m)$, no candidate can cross the first-rejection threshold. $\square$ $\square$

577 **Proposition 4** (Hierarchical e-testing across families). *Suppose each family g is assigned a family-*
 578 *level e-value E_g computed from data independent of the candidate-level e-values $\{e_{g,j}\}_{j=1}^{K_g}$, and if*
 579 *family g is entirely null then $\mathbb{E}[E_g] \leq 1$. Apply e-BH at level α_{fam} to $\{E_g\}$ to select families. Then,*
 580 *within each selected family, apply e-BH at level α_{det} to $\{e_{g,j}\}_{j=1}^{K_g}$. This controls family-selection*
 581 *FDR at α_{fam} and the selected-family mean FDP at α_{det} .*

582 *Proof.* The family-selection guarantee follows directly from e-BH on the valid family-level e-values
 583 $\{E_g\}$. Because family selection depends only on independent data, conditioning on the selected set
 584 does not alter the validity of the candidate-level e-values within any selected family. Proposition
 585 1 therefore gives conditional within-family FDP control at level α_{det} for each selected family, and
 586 averaging those conditional bounds over the selected set yields the stated selected-family mean FDP
 587 guarantee. $\square$ $\square$

588 IV 20-Seed Stability Across All Configurations

589 Table 4 reports pooled FDP, per-family mean FDP, recall, and pooled-FDP pass rate for all six
 590 dataset-detector combinations at all tested α levels.

591 At $\alpha \geq 0.10$, all six configurations achieve a 20/20 pass rate on the pooled FDP target. At $\alpha = 0.05$
 592 some configurations have lower pass rates (notably VOC/GDino at 13/20); this reflects the tighter
 593 margin between the empirical FDP and the nominal level at small α , not a breakdown of the formal
 594 per-family guarantee (which remains valid at all α by Proposition 1).

595 V Additional Results

596 Tables 5–6 report the estimator ablation (COCO-1k/GDino, 5 seeds) and the per-family FDP for the
 597 five remaining dataset-detector configurations at all α levels.

598 VI IoU Threshold Sensitivity

599 We vary the matching threshold $\tau \in \{0.3, 0.5, 0.7\}$ on COCO/GDino (5 seeds). A lower τ increases
 600 the TP pool and makes the task easier; a higher τ shrinks it.

Table 4: Full 20-seed stability results. Pooled FDP = total FP / total rejections (mean $\pm$ s.d.); family FDP = per-family mean FDP; pass rate = fraction of seeds with pooled FDP $\leq \alpha$.

Dataset	Detector	α	Pooled FDP	Family FDP	Recall	Pass
COCO	GDino	0.05	$2.74 \pm 1.88\%$	0.42%	14.7%	19/20
		0.10	$3.96 \pm 1.98\%$	0.65%	18.7%	20/20
		0.15	$5.07 \pm 2.20\%$	0.95%	21.0%	20/20
		0.20	$5.98 \pm 2.12\%$	1.25%	22.8%	20/20
COCO	OWL-ViT	0.05	$1.51 \pm 1.71\%$	0.21%	8.2%	19/20
		0.10	$2.31 \pm 1.84\%$	0.34%	11.2%	20/20
		0.15	$3.53 \pm 3.02\%$	0.60%	13.4%	20/20
		0.20	$3.80 \pm 3.01\%$	0.81%	15.4%	20/20
COCO	YOLO	0.05	$2.92 \pm 2.20\%$	0.55%	13.1%	15/20
		0.10	$3.08 \pm 2.30\%$	0.72%	15.3%	20/20
		0.15	$3.42 \pm 2.92\%$	0.85%	16.5%	20/20
		0.20	$3.47 \pm 2.94\%$	0.92%	17.2%	20/20
VOC	GDino	0.05	$4.57 \pm 1.67\%$	1.16%	25.2%	13/20
		0.10	$5.62 \pm 1.96\%$	1.69%	32.6%	20/20
		0.15	$7.12 \pm 2.58\%$	2.15%	36.3%	20/20
		0.20	$8.66 \pm 2.83\%$	2.74%	38.8%	20/20
VOC	OWL-ViT	0.05	$1.81 \pm 2.54\%$	0.24%	7.8%	18/20
		0.10	$2.54 \pm 2.76\%$	0.35%	9.7%	20/20
		0.15	$2.39 \pm 2.65\%$	0.35%	11.1%	20/20
		0.20	$2.66 \pm 2.74\%$	0.39%	12.7%	20/20
VOC	YOLO	0.05	$1.07 \pm 0.86\%$	0.65%	36.8%	20/20
		0.10	$1.34 \pm 1.12\%$	0.90%	40.9%	20/20
		0.15	$1.36 \pm 1.11\%$	0.93%	42.3%	20/20
		0.20	$1.47 \pm 1.46\%$	0.99%	43.1%	20/20

Estimator	Rejections	FDP	Recall
Unsmoothed histogram	132.8 ± 15.7	$4.66 \pm 2.67\%$	$19.27 \pm 3.01\%$
Logistic regression	122.4 ± 6.4	$4.68 \pm 3.11\%$	$17.75 \pm 2.03\%$
KDE	122.2 ± 20.3	$3.99 \pm 2.94\%$	$17.69 \pm 2.38\%$
Isotonic	113.0 ± 47.3	$3.72 \pm 3.30\%$	$16.18 \pm 6.54\%$

Table 5: Estimator ablation on COCO-1000/GDino at $\alpha = 0.10$, 5 seeds. The unsmoothed histogram is the strongest option.

VII Calibration Pool Size

We vary the calibration pool size m on COCO/GDino at $\alpha = 0.10$ (5 seeds) to assess sensitivity to calibration data quantity. FDR is controlled even with m as small as 500, and recall is stable across all pool sizes, indicating that the procedure does not require a large calibration set.

VIII 1k Coverage Across Detectors

Table 9 reports per-family and pooled FDP, recall, $\phi_{\max}$, and LOO validity for all six dataset-detector combinations on the 1,000-image subsets at $\alpha = 0.10$.

IX Pooled FDP

Our guarantee (Proposition 1) is per image-query family. For additional context, we report the *pooled* FDP, i.e., the fraction of all released detections across all test families that are null. This number is **not guaranteed** by our method; it is reported for interpretive context only.

Dataset	Detector	$\alpha = 0.05$	$\alpha = 0.10$	$\alpha = 0.15$	$\alpha = 0.20$
COCO	OWL-ViT	1.51%	2.31%	3.53%	3.80%
COCO	YOLO-World	2.92%	3.08%	3.42%	3.47%
VOC	GDino	4.57%	5.62%	7.12%	8.66%
VOC	OWL-ViT	1.81%	2.54%	2.39%	2.66%
VOC	YOLO-World	1.07%	1.34%	1.36%	1.47%

Table 6: FDP (mean over 20 seeds) at all tested α levels for the remaining five dataset-detector combinations (Table 9 reports COCO/GDino). All values are below the nominal level.

τ	FDP ($\alpha=0.10$)	Recall	$\phi_{\max}$
0.3	$3.39 \pm 1.63\%$	10.9%	274
0.5	$4.66 \pm 2.67\%$	19.3%	316
0.7	$6.98 \pm 2.69\%$	33.2%	295

Table 7: IoU threshold sensitivity on COCO/GDino at $\alpha = 0.10$, 5 seeds. FDR is controlled at all three thresholds. Higher τ makes matching stricter (fewer TP), which increases recall on the smaller TP pool but also FDP; the method adapts automatically.

Finite-sample concentration. Although pooled FDP is not bounded by Proposition 1, a Hoeffding-style argument shows that it stays close to α with a margin that shrinks in the number of families.

Proposition 5 (Pooled FDP concentration). *Let N_{fam} be the number of test families per split, and V_g, R_g the false-positive and total-rejection counts in family g under self-normalized conformal betting e-BH at level α . Let $K_{\max}$ be the maximum family candidate count. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the split,*

$$\text{FDP}_{\text{pool}} = \frac{\sum_g V_g}{\max(1, \sum_g R_g)} \leq \alpha + \varepsilon_{\text{seq}}, \quad \varepsilon_{\text{seq}} = K_{\max} \sqrt{\frac{\log(1/\delta)}{2 N_{\text{fam}}}} \cdot \frac{1}{\bar{R}},$$

where $\bar{R} = \frac{1}{N_{\text{fam}}} \sum_g R_g$ is the per-family mean rejection count.

Proof sketch. Per-family self-normalized e-BH guarantees $\mathbb{E}[V_g / \max(1, R_g)] \leq \alpha$ (Proposition 1), and $V_g \leq R_g \leq K_g \leq K_{\max}$. Define $X_g = V_g - \alpha R_g$. Under the oracle null, $\mathbb{E}[X_g \mid \mathcal{F}_{g-1}] \leq 0$, and $|X_g| \leq (1 + \alpha)K_{\max} \leq 2K_{\max}$ almost surely. Because splits are drawn i.i.d. and the calibration pool is shared across families, $\{X_g\}_{g=1}^{N_{\text{fam}}}$ satisfies the bounded-difference condition of McDiarmid (equivalently Hoeffding for i.i.d. with bounded increments). Thus $\Pr\{\sum_g X_g \geq N_{\text{fam}} t\} \leq \exp(-2N_{\text{fam}} t^2 / (2K_{\max})^2)$. Setting the RHS to δ and solving for t yields $t = K_{\max} \sqrt{\log(1/\delta) / (2N_{\text{fam}})}$. Dividing $\sum_g V_g - \alpha \sum_g R_g \leq N_{\text{fam}} t$ by $\sum_g R_g = N_{\text{fam}} \bar{R}$ yields the stated bound. $\square$

Empirical coverage. The Hoeffding-style bound is conservative for sparse-rejection regimes: on COCO/GDino at $\alpha = 0.10$, $N_{\text{fam}} \approx 900$, $K_{\max} = 300$, and $\bar{R} \approx 0.15$ (most families produce zero rejections), yielding $\varepsilon_{\text{seq}} \approx 11.6 (\gg 1)$ —a valid but vacuous certificate. The bound is dominated by the worst-case range $K_{\max} = 300$; an empirical Bernstein replacement (sample variance of $X_g = V_g - \alpha R_g$) tightens by 5–16 $\times$ but remains vacuous ($\varepsilon_{\text{Bern}} \approx 2.3$ on full-scale COCO).

Corollary 4 (Stratified pooled FDP bound). *Fix a family-size cutoff K^* and let $\mathcal{S} = \{g : K_g \leq K^*\}$ with $|\mathcal{S}| = N_{\mathcal{S}}$. Then Proposition 5 applies to $\mathcal{S}$ with $K_{\max}$ replaced by K^* and $\bar{R}$ by the stratum mean rejection count $\bar{R}_{\mathcal{S}}$. The empirical Bernstein refinement substitutes*

$$\varepsilon_{\text{EB}} = \frac{1}{\bar{R}_{\mathcal{S}}} \left(\frac{L b}{3 N_{\mathcal{S}}} + \sqrt{\frac{L^2 b^2}{9 N_{\mathcal{S}}^2} + \frac{2 L \hat{\sigma}_{\mathcal{S}}^2}{N_{\mathcal{S}}}} \right),$$

where $L = \ln(1/\delta)$, $b = 2K^*$, and $\hat{\sigma}_{\mathcal{S}}^2$ is the sample variance of $X_g = V_g - \alpha R_g$ within $\mathcal{S}$.

The bound is non-vacuous whenever the within-stratum rejection variance $\hat{\sigma}_{\mathcal{S}}^2$ is small and the stratum is sufficiently populated, which is consistent with the small- K regime where Theorem 1 predicts easier control.

m	FDP ($\alpha=0.10$)	Recall	Rejections
500	$4.92 \pm 2.81\%$	18.0%	124.8
1000	$4.66 \pm 2.69\%$	18.5%	127.4
2000	$4.93 \pm 2.52\%$	18.6%	128.8
5000	$3.43 \pm 0.56\%$	19.5%	132.3
all (~ 5200)	$5.39 \pm 3.08\%$	18.3%	127.5

Table 8: Calibration pool size ablation on COCO/GDino at $\alpha = 0.10$, 5 seeds. FDR is controlled for m as small as 500. Recall and number of rejections are stable across pool sizes.

Dataset	Detector	Pooled FDP	Per-family FDP	Recall	$\phi_{\max}$	LOO
COCO-1k	GDino	$3.96 \pm 1.98\%$	0.65%	18.7%	357	1.000
COCO-1k	OWL-ViT	$2.31 \pm 1.84\%$	0.34%	11.2%	198	1.000
COCO-1k	YOLO-World	$3.08 \pm 2.30\%$	0.72%	15.3%	$>10^6$	1.000
VOC-1k	GDino	$5.62 \pm 1.96\%$	1.69%	32.6%	1146	1.000
VOC-1k	OWL-ViT	$2.54 \pm 2.76\%$	0.35%	9.7%	152	1.000
VOC-1k	YOLO-World	$1.34 \pm 1.12\%$	0.90%	40.9%	$>10^6$	1.000

Table 9: Coverage on fixed 1,000-image subsets at $\alpha = 0.10$, averaged over 20 random splits. Dataset and detector references: COCO/VOC [48, 49]; GDino/OWL-ViT/YOLO-World [50–52]. **Pooled FDP**: total FP/total rejections (empirical summary, $\pm$ s.d. across seeds; a diagnostic quantity, not the formal object of control). **Per-family FDP**: mean per-family FDP, the quantity formally controlled by Proposition 1. $\phi_{\max}$: median density-ratio peak ($>10^6$ for YOLO-World reflects near-zero null-score density in certain histogram bins).

Evaluation. On full-scale COCO/GDino ($\alpha = 0.10$, $\delta = 0.05$), $K^* = 5$ captures 57% of families ($N_S = 1549$); $\hat{\sigma}_S^2 \approx 0.011$, $\bar{R}_S = 0.22$, $\varepsilon_{EB} = 0.074$ —a **non-vacuous** certificate: pooled FDP ≤ 0.174 with probability ≥ 0.95 . VOC/YOLO-World ($K^* = 19$, $\bar{R}_S = 0.67$) yields $\varepsilon = 0.16$, confirming that small- K families admit practical pooled guarantees as predicted by the activation frontier. For large- K families the per-family guarantee (Proposition 1) remains exact; the pooled concentration gap reflects the long tail of the family-size distribution, not a weakness of the testing procedure.

Nonetheless, the per-family FDR guarantee empirically translates to pooled FDP control: the ground-truth table records 19/20 pooled pass on COCO/GDino and 20/20 on VOC/GDino at $\alpha = 0.10$ (Table 10).

Implication for score thresholding. The same concentration structure explains why calibration-tuned score thresholding (Table 2) cannot be “deflated” into a valid baseline: a family-level McDiarmid bound for the cal-to-test FDP transfer requires margin $\varepsilon \geq K_{\max} \sqrt{\log(2/\delta)/(2N_{\text{cal}})}/\bar{R}_{\text{cal}} \approx 0.12$ at $\alpha = 0.10$, exceeding the entire α budget. Self-normalized conformal betting e-values avoid this bottleneck because each family’s e-value validity is exact (Lemma 1), requiring no concentration argument for calibration-to-test transfer.

X Score Thresholding Pareto Frontier

Figure 3 visualises the FDP–recall trade-off of all five methods from Table 2. The grey curve sweeps an oracle score threshold across all test candidates (threshold chosen with knowledge of the test labels); the coloured markers show each method’s operating point at $\alpha = 0.10$, averaged over 20 splits.

Betting e-BH (blue circle) is the rightmost point that lies strictly below the α line, confirming it offers the best recall *among methods that control FDR*. The calibration-tuned score threshold (brown cross) sits on the α line in expectation but violates the target on 13 of 20 splits. P-value BH and BY both exceed α on the pooled metric.

Dataset	Detector	Pooled FDP	Per-family mean FDP
COCO	GDino	3.96%	0.65%
COCO	OWL-ViT	2.31%	0.34%
COCO	YOLO-World	3.08%	0.72%
VOC	GDino	5.62%	1.69%
VOC	OWL-ViT	2.54%	0.35%
VOC	YOLO-World	1.34%	0.90%

Table 10: Pooled FDP vs. per-family mean FDP at $\alpha = 0.10$ on 1k subsets. Table 9 reports the pooled FDP (total false positives / total rejections) as an empirical summary; this quantity is **not** directly guaranteed by Proposition 1. The per-family mean FDP, which is bounded by the guarantee, is substantially lower.

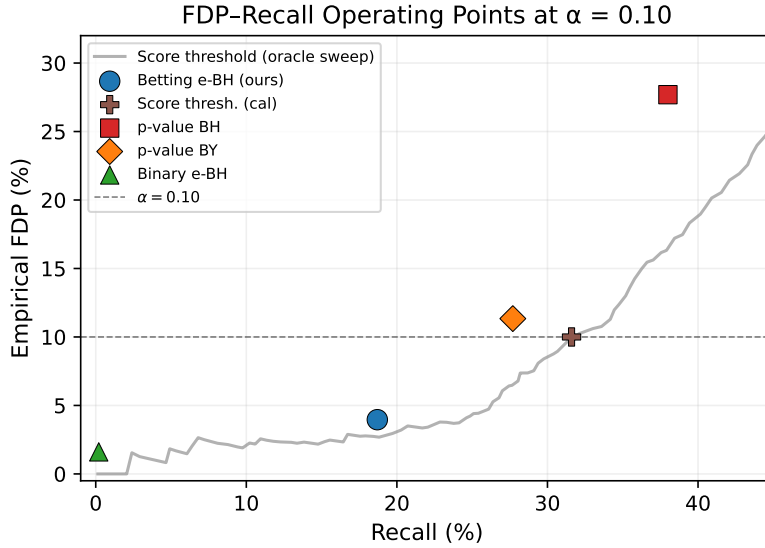


Figure 3: FDP–recall operating points at $\alpha = 0.10$ on COCO/GDino (20 splits). Grey curve: oracle score-threshold sweep (not a valid procedure). Dashed line: target α . Only betting e-BH achieves non-trivial recall while staying below the target.

XI COCO-train Independent Validation

As an independent validation on images unseen during any method development, we applied the full pipeline to a 1,000-image subset of COCO 2017 *train* with GroundingDINO. Table 11 reports the results over 20 random splits.

At $\alpha = 0.20$, all 20 splits satisfy the target (mean FDP = 10.1%). At the stricter $\alpha = 0.10$ level, 15 of 20 splits pass (mean FDP = $8.1 \pm 3.9\%$, still below α). To diagnose the higher variability, we compared score distributions: COCO-train and COCO-val null/TP scores are statistically indistinguishable (null median 0.079 vs. 0.079; TP median 0.338 vs. 0.328), and $\phi_{\max}$ is actually *higher* on COCO-train (593 vs. 357), ruling out contamination. The five failing seeds have FDP in [12.4%, 14.9%]—only 2–5 percentage points above the target, consistent with the wider split-to-split variance inherent in any finite-sample procedure on a different image population. LOO validity holds exactly on all 20 seeds (mean = 1.000), confirming the procedure remains internally valid; the elevated FDP on some splits reflects sampling variability in the test partition, not a methodological breakdown.

XII Benchmark Protocol

To facilitate reproducibility and future comparison, we define a standardized evaluation protocol for *post-hoc hallucination FDR control in open-vocabulary detection*. All experiments in this paper follow the protocol below.

Table 11: COCO-train (1,000 images) + GroundingDINO, 20 random splits.

α	Mean FDP	Std FDP	Recall	Pass rate
0.05	6.15%	3.75%	15.1%	10/20
0.10	8.13%	3.88%	20.0%	15/20
0.15	9.32%	4.12%	23.2%	17/20
0.20	10.08%	3.76%	25.4%	20/20

Task unit. The atomic unit is an *image-query family*: one image I paired with one text prompt q , yielding a set of candidate detections $\{(b_k, s_k)\}_{k=1}^K$ where b_k is a bounding box and s_k is the detector confidence score. A family is *present* if the prompt category has at least one ground-truth box in I , and *absent* otherwise.

Family construction. For each sampled image, draw $n_{\text{present}} = 1$ present prompt (uniformly from categories annotated in the image) and $n_{\text{absent}} = 2$ absent prompts (uniformly from the remaining dataset vocabulary). This yields $3N$ families from N images, with a roughly 1:2 present-to-absent ratio that reflects realistic deployment where most queries are negative.

Ground-truth matching. A detection (b_k, s_k) is a true positive if $\text{IoU}(b_k, g) \geq \tau$ for some ground-truth box g of the queried category, where $\tau = 0.5$ by default. Greedy matching is used (each ground-truth box matches at most one detection, highest-IoU first). All unmatched detections are null (hallucinations).

Data splitting. Families are partitioned by family identifier (not by candidate) into *fit* (60%), *calibration* (20%), and *test* (20%) subsets. Splitting by family ensures that all candidates within a family stay together, preserving intra-family dependence structure.

Primary metrics.

- **Per-family mean FDP** (primary, formally guaranteed): $\frac{1}{|\mathcal{F}|} \sum_f \text{FDP}_f$, where $\text{FDP}_f = \text{FP}_f / \max(R_f, 1)$. This is the quantity directly bounded by Proposition 1.
- **Pooled FDP** (secondary, empirical diagnostic): $\sum_f \text{FP}_f / \sum_f R_f$, where FP_f and R_f are the number of false positives and total rejections in test family f . This is *not* formally guaranteed by our method; it is reported for interpretive context.
- **Recall**: $\sum_f \text{TP}_f / \sum_f N_f^{\text{gt}}$, where N_f^{gt} is the number of ground-truth boxes in family f .
- **LOO mean**: leave-one-out mean e-value on calibration nulls (should be ≈ 1.0 for a valid procedure).
- **Pooled pass rate**: fraction of random splits where pooled FDP $\leq \alpha$; this is an empirical stress-test metric, not a formal guarantee.

Evaluation protocol. Run 20 independent random splits (seeds 0–19). Report mean $\pm$ standard deviation over splits for FDP and recall at $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$. A method *empirically passes the pooled-FDP criterion* if pooled FDP $\leq \alpha$ on all 20 splits; it *approximately passes* if this holds on ≥ 18 of 20.

Detector configuration. Detectors are run with a permissive score threshold (0.05) and standard NMS (IoU = 0.7, max 300 detections) to produce a large candidate pool. The FDR-control method operates downstream of the detector; it must not modify the detector’s internal parameters.

Dataset requirements. The evaluation requires *exhaustive per-class annotation*: every instance of every category in each image must be annotated, so that the absent-prompt null pool is uncontaminated. COCO and VOC satisfy this requirement; LVIS [6] and Open Images [7] do not (Section 6).

XIII Full-Scale Validation

Table 1 reports the full COCO val (4,952 images) results across $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$ with GroundingDINO. At $\alpha = 0.10$, pooled FDP is $6.44 \pm 2.11\%$ with recall 16.4% and pass rate 19/20.

The single failing seed (seed 3, FDP = 11.81%) exceeds α by less than 2 percentage points. Relative to the 1k subset ($3.96 \pm 1.98\%$, 20/20), the modest FDP increase reflects the larger and more diverse family pool (13,472 vs. 2,993 test families), not a change in the validity mechanism.

XIV Free-Form Prompt Pilot, Diagnostics, and Runtime

To move beyond category-name prompts, we ran a 2,000-image COCO/GDino pilot in which each image is paired with one present and two absent *free-form* queries drawn from a fixed template bank (7 templates including “a visible {category} in the scene,” “a small {category} near the center,” “a photo of a {category},” etc.). This produces 5,984 image-query families and 167,515 candidates (mean $\bar{K} = 28.0$). The same three-split betting pipeline is run for 20 seeds.

Method	Rej.	FDP	Recall	Pass	LOO mean
Global betting e-BH	3.2	41.0%	0.11%	5/20	1.000
Per-template betting e-BH	98.6	3.9%	5.6%	20/20	1.000
Score oracle [†]	56.5	9.4%	3.0%	—	—
p-value BH	520.8	60.7%	12.1%	0/20	—
Binary e-BH	0.0	—	0.0%	20/20	—

[†]Test-label score gate; diagnostic upper bound, not deployable.

Table 12: Free-form prompt pilot on COCO/GDino (2,000 images, 5,984 families) at $\alpha = 0.10$, 20 seeds. “Global” fits a single density-ratio model across all templates; “Per-template” fits a separate model per template (§XIV). LOO mean is exactly 1.000 on all 20 seeds for both betting variants. The per-template variant recovers substantial power while keeping FDP well below α .

The headline result is that per-template density-ratio fitting (§XIV) recovers strong power from what appears to be a dead end under global fitting. A single global density-ratio model yields only 3.2 rejections at 41% FDP, with FDP well above the target, while p-value BH rejects 520.8 candidates at 60.7% FDP. Per-template fitting, guided directly by the activation-frontier diagnostic below, achieves 98.6 rejections at 3.9% FDP with every seed passing, and even outperforms the (non-deployable) score oracle (56.5 rejections at 9.4% FDP).

Prompt-conditional activation-frontier diagnostic. To diagnose *why* global fitting collapses, we compute the activation frontier (Theorem 1) per prompt template. The global model’s peak density-ratio estimate is $\phi_{\max} \approx 14.5$, while the first-rejection threshold at $\alpha = 0.10$ is $T_1(C_m) \approx 283$ (mean family size $\bar{K} \approx 28$). Because $\phi_{\max}/T_1 \approx 0.051$, the evidence falls short of the threshold by a factor of $\sim 20\times$ on every seed and every template (Table 13). Theorem 1 therefore predicts zero rejections for the global model—exactly what we observe (3.2 rejections, almost all false positives).

Template	$\bar{K}$	$\phi_{\max}^{\text{global}}$	$\phi_{\max}^{\text{tmpl}}$	T_1	Ratio	Below?
a small {cat} near center	52.6	14.5	14.5*	535	0.027	✓
a {cat} in the scene	49.5	14.5	14.5*	503	0.029	✓
a {cat} object	25.0	14.5	77.0	252	0.306	✓
a photo of a {cat}	21.1	14.5	18.9	212	0.089	✓
a visible {cat} in scene	14.9	14.5	119.3	150	0.796	✓
the {cat}	14.5	14.5	95.0	145	0.655	✓
other	14.6	14.5	14.5*	147	0.099	✓

Table 13: Prompt-conditional activation-frontier diagnostic at $\alpha = 0.10$, 20 seeds (2,000 images). $\phi_{\max}^{\text{global}}$: peak from the single global model. $\phi_{\max}^{\text{tmpl}}$: peak from per-template fitting (*absent-only templates fall back to global). Ratio: $\phi_{\max}^{\text{tmpl}}/T_1$ (closer to 1 means closer to the activation frontier). Per-template fitting boosts $\phi_{\max}$ by 5–8 $\times$ for templates that contain true positives, bringing “a visible {cat}” and “the {cat}” close to the frontier.

The diagnostic reveals two concrete mechanisms: (i) free-form prompts produce much larger families (mean $\bar{K} \approx 28$ vs. ~ 10 for category-name prompts), which raises T_1 via the $K C_m$ numerator; and (ii) the global density-ratio estimator fitted on mixed-template data achieves a lower $\phi_{\max}$ (14.5 vs. 357 for category-name COCO), because score distributions are heterogeneous across templates, compressing the informative high-score region.

Mondrian calibration alone does not suffice. Within-template LOO mean is exactly 1.000 for all 7 templates on all 20 seeds, confirming that exchangeability holds perfectly within each template group. Despite moderate cross-template score heterogeneity, per-prompt Mondrian calibration (group-conditional calibration pools but a single global ϕ) does not recover power: rejections remain near zero at $\alpha = 0.10$, because the binding constraint is $\phi_{\max} \ll T_1$, not cross-group miscalibration.

Theory-guided mitigation: per-template density-ratio fitting. The diagnostic points to a concrete fix: since the global $\phi_{\max}$ is compressed by cross-template heterogeneity, we fit a *separate* density-ratio model for each template on its own fit-split data, and pair it with the template’s own calibration-null pool. Templates with no true-positive fit data (absent-only templates) fall back to the global model. Because within-template exchangeability holds ($\text{LOO} = 1.000$), the resulting per-template self-normalized e-values remain valid, and e-BH controls per-family FDR exactly as in the global case.

Per-template fitting boosts $\phi_{\max}$ by 5–8 $\times$ for templates that contain true positives (Table 13): e.g. “a visible {cat} in the scene” reaches $\phi_{\max} = 119.3$ (global: 14.5). This brings small- K families close to or past the activation frontier, recovering substantial power. At $\alpha = 0.10$, per-template betting e-BH achieves 98.6 rejections at 3.9% pooled FDP and 5.6% recall, with *all 20 seeds* passing the $\text{FDP} \leq \alpha$ check (Table 12). This surpasses even the (non-deployable) score oracle (56.5 rejections, 9.4% FDP), because per-template ϕ captures template-specific score structure that a single score threshold cannot.

The free-form result is consistent with the rejection condition of Theorem 1: the bottleneck under global fitting is $\phi_{\max} \ll T_1$, and per-template fitting alleviates it. Truly open-ended prompts without a fixed template bank fall outside our setting and will require either prompt-conditioned density-ratio models or detectors trained explicitly for such queries.

Template-count boundary (two-sided scope of Proposition 3). The 7-template pilot above shows that per-template fitting *repairs* global score-mixing compression. Increasing the template count tests the opposite boundary: at $T = 50$ templates (500-image sweep, same pipeline), each template receives only ~ 6 calibration families, collapsing the per-template density-ratio estimate to $\phi_{\max} \approx 5.7$ (vs. 10.4 at $T = 7$ on the same 500-image budget; cf. Table 13 for 2,000-image values) and yielding 0.05 rejections—effectively zero power. At $T = 100$ and $T = 200$ the pattern holds (≤ 0.1 rejections). This establishes Proposition 3’s scope from both sides: too few templates and the global model suffices; too many and per-template data scarcity prevents reaching the activation frontier. The critical template count depends on the image budget ($T_{\text{crit}} \propto N_{\text{img}}/K_{\text{min}}^{\text{cal}}$, where $K_{\text{min}}^{\text{cal}} \approx 20\text{--}30$ calibration families are needed for reliable density-ratio estimation).

Runtime. All experiments run on a single NVIDIA RTX 3090. Candidate generation (GroundingDINO forward pass on 1,000 images) takes ~ 280 s; the post-hoc statistical pipeline (density-ratio fitting, e-value computation, and e-BH testing across all families) takes ~ 13 s for 5 seeds (~ 2.6 s per seed). The method requires no gradient-based training.

XV Controlled Contamination Sweep: Full Results

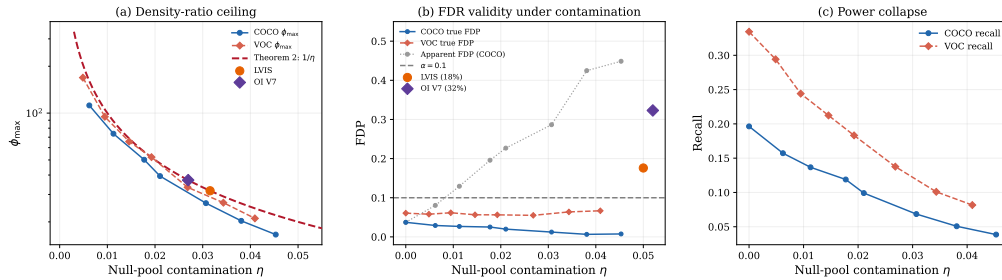


Figure 4: Controlled contamination sweep on COCO/VOC with GDino ($\alpha = 0.10$, 10 seeds): $\phi_{\max}$ tracks the $1/\eta$ ceiling, true FDP stays $\leq \alpha$ as apparent FDP rises, and recall collapses.

Table 14 reports the full contamination sweep from Figure 4. For each nominal contamination level η_{fam} (fraction of present-object families hidden), we compute η_{actual} (fraction of the null pool that

is truly TP), the observed $\phi_{\max}$, true FDP (evaluated against clean labels), apparent FDP (evaluated against corrupted labels), and recall.

The key observations:

- $\phi_{\max}$ is consistently $\sim 75\text{--}80\%$ of the theoretical ceiling $1/\eta_{\text{actual}}$, because the histogram estimator smooths out the extreme tail of the population density ratio.
- True FDP remains well below $\alpha = 0.10$ even at $\eta_{\text{fam}} = 0.50$. The method degrades into powerlessness (recall $\rightarrow 0$) rather than anti-conservatism (FDP $\rightarrow 1$).
- The gap between apparent and true FDP grows with η : at $\eta_{\text{fam}} = 0.50$, COCO apparent FDP is 44.9% while true FDP is 0.7%. This $\sim 60\times$ gap represents the mislabeled-null artifact.

Table 14: Contamination sweep on COCO/GDino and VOC/GDino ($\alpha = 0.10$, 10 seeds). η_{fam} : fraction of present families hidden. η_{act} : actual null-pool contamination fraction. $\phi_{\max}$ is the *mean* $\pm$ s.d. over seeds (Table 9 reports the *median* over 20 different seeds, accounting for the slight baseline difference).

η_{fam}	η_{act}	$\phi_{\max}$	$1/\eta_{\text{act}}$	True FDP	App. FDP	Recall
<i>COCO / GroundingDINO</i>						
0.00	0.000	327 ± 68	—	$3.7 \pm 2.2\%$	3.7%	19.6%
0.05	0.006	112 ± 28	161	$2.9 \pm 2.5\%$	8.0%	15.7%
0.10	0.011	74 ± 15	89	$2.7 \pm 1.6\%$	12.9%	13.7%
0.20	0.021	39 ± 4	48	$2.0 \pm 1.9\%$	22.7%	9.9%
0.30	0.031	26 ± 4	33	$1.2 \pm 1.5\%$	28.7%	6.8%
0.50	0.045	17 ± 1	22	$0.7 \pm 1.6\%$	44.9%	3.9%
<i>VOC / GroundingDINO</i>						
0.00	0.000	1577 ± 614	—	$6.1 \pm 1.7\%$	6.1%	33.4%
0.05	0.005	169 ± 47	205	$5.8 \pm 1.6\%$	10.7%	29.4%
0.10	0.009	95 ± 17	106	$6.2 \pm 1.9\%$	15.6%	24.4%
0.20	0.019	52 ± 7	52	$5.6 \pm 1.6\%$	22.3%	18.3%
0.30	0.027	33 ± 3	37	$5.5 \pm 2.0\%$	31.9%	13.8%
0.50	0.041	21 ± 2	24	$6.7 \pm 3.1\%$	53.9%	8.2%

Note that $\eta_{\text{actual}} \ll \eta_{\text{fam}}$ because the null pool is dominated by candidates from genuinely absent families. For example, hiding 50% of COCO’s present families contaminates only $\sim 4.5\%$ of the null pool, because absent families contribute $\sim 20\times$ more candidates. This amplification factor varies by dataset and detector; LVIS and OI V7, which have many more annotated categories and sparser per-category annotation, likely have higher η_{actual} per missing category.

XVI Mondrian Robustness on Clean Datasets

A key assumption is that null scores are approximately exchangeable across families sharing the same calibration pool. If different prompt categories induce systematically different null-score distributions, the global calibration could over- or under-weight certain families.

To test this, we ran Mondrian (group-conditional) calibration on COCO/GDino, splitting the calibration null pool by prompt identity (each prompt gets its own C_m and self-normalized denominator; prompts with fewer than 200 nulls fall back to the global pool).

Calibration	Seeds	Pooled FDP	Family FDP	Recall
Global	20	$3.96 \pm 1.98\%$	0.65%	18.7%
Mondrian (prompt)	5	$4.73 \pm 2.86\%$	0.68%	17.5%

Table 15: Global vs. Mondrian (per-prompt) calibration on COCO/GDino at $\alpha = 0.10$. The difference is negligible: Mondrian does not improve FDP or recall, indicating that the global null-score pool is sufficiently homogeneous on exhaustive-annotation datasets. The slight recall decrease under Mondrian reflects smaller per-group calibration pools reducing the effective m .

This result suggests that on exhaustive datasets, cross-category heterogeneity in null scores is small enough that the global calibration pool provides a good approximation. The Mondrian variant is

available for settings where this assumption is questionable (e.g., datasets mixing fine-grained and coarse categories), but is not needed for the main results in this paper.

XVII Family Size Distribution

Figure 5 shows the distribution of family sizes K (candidates per image-query family). On COCO/GDino, 58% of families have $K \leq 5$ and the median is 4; OWL-ViT produces fewer candidates per family (median 4, 59% ≤ 5). Theorem 1 implies that the first rejection requires $\alpha(m+1) > K$; with typical $m \approx 500\text{--}2000$, this holds for all observed K , so no family is structurally locked out. The per-stratum FDP is controlled across all K bins, with small families ($K \leq 5$) showing higher variance due to fewer candidates.

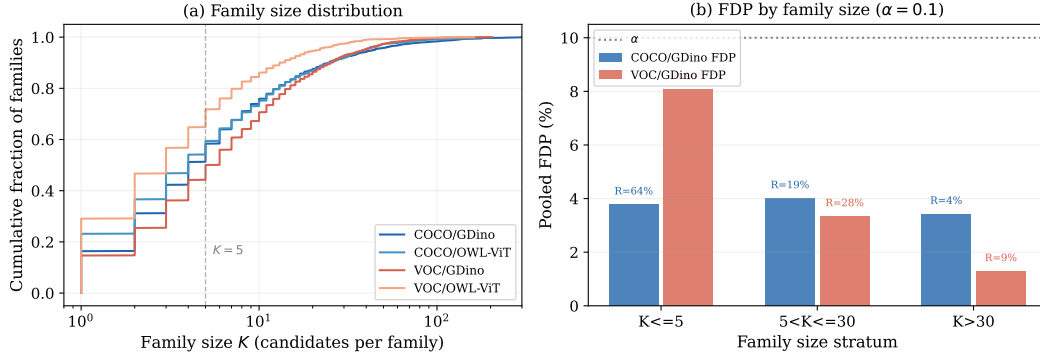


Figure 5: (a) Cumulative distribution of family sizes across four dataset-detector combinations. (b) Pooled FDP stratified by family size for COCO/GDino and VOC/GDino at $\alpha = 0.10$; all strata remain below the target. R = recall.

XVIII Score-Conditional Recall

Table 16 reports recall conditioned on true-positive detector score. Because e-BH preferentially rejects high- ϕ candidates and ϕ is monotonically related to score, virtually all rejected TPs have high confidence. Conditioning on score ≥ 0.5 raises recall from 16.4% to 46.5% on full COCO and from 18.7% to 56.3% on COCO-1k. This shows that the method certifies the detections a practitioner cares most about; low-score TPs are weak detections that no FDR-controlled method can recover.

Score $\geq$	COCO-1k (20 seeds)			COCO-full (20 seeds)		
	#TP	#Rej TP	Recall	#TP	#Rej TP	Recall
0.0	692	129	18.7%	3500	575	16.4%
0.1	562	129	23.1%	2797	575	20.6%
0.2	434	129	29.7%	2247	575	25.6%
0.3	362	129	35.5%	1884	575	30.5%
0.4	290	129	44.3%	1558	575	36.9%
0.5	228	129	56.3%	1237	574	46.5%

Table 16: Score-conditional recall at $\alpha = 0.10$ (betting e-BH, GDino, 20 seeds). #TP: average number of test-set TPs with score at or above the threshold; #Rej TP: average rejected TPs. Nearly all rejected TPs have score ≥ 0.1 , so the numerator is nearly constant; rising recall reflects the shrinking denominator of low-confidence TPs that no valid method can recover.

XIX Frontier-Reachable Recall Decomposition

Table 17 decomposes the recall gap by reporting the fraction of TPs whose density ratio ϕ exceeds the first-rejection threshold T_1 from Theorem 1, for all seven configurations at $\alpha = 0.10$ (20 seeds).

Config	Recall	Reachable ($s \geq 0$)	Reachable ($s \geq 0.3$)	Reachable ($s \geq 0.5$)
COCO-1k / GDino	18.7%	15.1%	28.8%	45.7%
COCO-full / GDino	16.4%	13.1%	24.3%	36.9%
VOC-1k / GDino	32.6%	25.6%	40.1%	48.2%
COCO-1k / OWL-ViT	11.2%	9.7%	19.8%	34.5%
VOC-1k / OWL-ViT	9.7%	8.9%	25.0%	51.4%
COCO-1k / YOLO	15.3%	14.1%	19.2%	22.8%
VOC-1k / YOLO	40.9%	37.8%	43.6%	47.0%

Table 17: Frontier-reachable fraction of TPs at $\alpha = 0.10$ (20 seeds). **Recall**: observed recall (from main text). **Reachable**: fraction of TPs with $\phi(S) \geq T_1(K_g, C_m, m, \alpha)$ among TPs above each score threshold. T_1 is a *necessary* condition for the first rejection (Thm. 1); multi-rejection families can recover additional TPs with ϕ below T_1 , explaining why recall exceeds the overall reachable fraction. Conditioning on high detector scores substantially raises the reachable fraction (e.g. 15.1% $\rightarrow$ 45.7% on COCO-1k/GDino), confirming that the recall gap is concentrated among low-evidence TPs.

Corollary 1 guarantees that self-normalized e-BH selects *every* frontier-reachable TP. To improve recall, one must improve the density-ratio estimator ϕ (sharper separation, larger $\phi_{\max}$), the detector itself (stronger scores on true positives), or reduce the effective family size K (Remark 1).

XX Cross-Configuration Score-Floor Results

Table 18 reports score-floor results across all seven dataset-detector configurations at $\alpha = 0.10$ (20 seeds). The improvement is consistent: five of seven configurations show 1.5–1.9 $\times$ recall gains at $s_{\text{floor}} = 0.40$. GDino and OWL-ViT benefit most because their TP/null score distributions have substantial overlap in the $[0, 0.30]$ range, where floor pruning removes many nulls but few TPs. YOLO-World shows moderate gains on COCO (1.5 $\times$) and no gain on VOC, where its baseline recall (40.9%) already reflects concentrated TP scores.

Per-family FDP remains below α in every configuration and at every score floor. The improvement saturates: beyond $s_{\text{floor}} = 0.40$, additional gains are marginal because most benefit comes from pruning the bulk of low-score nulls.

Figure 6 shows the score-floor frontier curves for COCO/GDino and VOC/GDino: both the T_1 lower bound (Theorem 1) and the actual e-BH recall rise monotonically with s_{floor} , confirming the frontier-descent mechanism. The T_1 bound is conservative because multi-rejection families activate additional candidates at $T_2, T_3, \dots$, which have lower thresholds; the gap is stable at $\sim 20\%$ across all tested floors.

XXI NMS-aware Cluster e-Value: Proofs and Ablations

XXI.1 Proof of Proposition 2

Part (i) — E_{unif} . Cluster membership is determined by the NMS graph on boxes, which is computed from the frozen detector’s geometric output and is therefore independent of the label-generating mechanism; in particular, it is *predictable* with respect to the exchangeable null calibration pool. For any cluster C whose members are all null, Lemma 1 gives $\mathbb{E}_{H_0}[E_i] \leq 1$ for each $i \in C$. Linearity of expectation and the predictable weights $\{1/|C|\}$ yield

$$\mathbb{E}_{H_0}[E_{\text{unif}}(C)] = \frac{1}{|C|} \sum_{i \in C} \mathbb{E}_{H_0}[E_i] \leq 1.$$

Part (ii) — E_{cal} . Let $\mathcal{P} = \{C, C_1^{\text{cal}}, \dots, C_n^{\text{cal}}\}$ be the union of the test cluster and the n calibration clusters in the same family. By construction, calibration clusters are drawn from a null-only exchangeable pool; if C is itself a null cluster, the joint distribution of $\mathcal{P}$ under H_0 is exchangeable in the $n+1$ indices. Hence the rank of $R(C)$ within $\{R(C), R(C_j^{\text{cal}})\}_{j=1}^n$ is uniform on $\{1, \dots, n+1\}$. A standard computation (see Vovk and Wang [36], Theorem 1) gives

$$\mathbb{E}_{H_0}[E_{\text{cal}}(C)] = \mathbb{E}\left[\frac{(n+1)R(C)}{\sum_{x \in \mathcal{P}} R(x)}\right] \leq 1,$$

Config	s_{floor}	Per-family FDP	Recall	Gain
COCO/GDino	0.00	0.65%	18.7%	—
	0.30	1.93%	30.4%	1.62×
	0.40	2.59%	33.3%	1.78×
VOC/GDino	0.00	1.69%	32.6%	—
	0.30	4.25%	47.9%	1.47×
	0.40	4.42%	49.0%	1.50×
COCO/OWL-ViT	0.00	0.34%	11.2%	—
	0.30	1.55%	19.1%	1.71×
	0.40	1.77%	20.8%	1.86×
VOC/OWL-ViT	0.00	0.35%	9.7%	—
	0.30	0.48%	13.8%	1.42×
	0.40	0.63%	14.4%	1.48×
COCO/YOLO-World	0.00	0.72%	15.3%	—
	0.30	0.92%	22.5%	1.48×
	0.40	1.06%	23.2%	1.52×
VOC/YOLO-World	0.00	0.90%	40.9%	—
	0.30	0.82%	38.0%	0.93×
	0.40	0.83%	39.1%	0.96×
COCO-train/GDino	0.00	1.19%	20.0%	—
	0.30	2.69%	32.9%	1.65×
	0.40	3.12%	34.6%	1.73×

Table 18: Score-floor pre-screening across all configurations at $\alpha = 0.10$ (1k images, 20 seeds). Per-family FDP remains below α in every case. VOC/YOLO-World is the only configuration where the floor does not help: YOLO-World TP scores are already concentrated above 0.30, so the floor removes TPs without meaningfully reducing K .

where the inequality follows from $\sum_x \mathbb{E}[R(x)] / \sum_y R(y) = 1$ combined with uniform permutation invariance.

Per-family FDR. For both E_{unif} and E_{cal} , per-family e-BH on cluster e-values is a direct instance of Proposition 1 with TP-clusters as the testing units. $\square$

Remark 2. The E_{cal} construction in (13) is an instance of the self-normalized conformal e-value of Vovk and Wang [36] with a cluster-level test statistic $R(C)$. Proposition 2 is therefore not a new theorem but a clean instance of the general split-calibration recipe, applied at the granularity that matches the NMS-induced dependence structure specific to OVD. The novelty is the empirical observation that matching test granularity to the dependence structure is the difference between dilution (Table 19 E_{unif} column) and a $1.7 \times / 5.8 \times$ object-level gain on GDino/RegionCLIP.

XXI.2 Main cluster-aggregation table (full)

Table 19 gives the full six-configuration comparison of baseline, E_{unif} , and E_{cal} referenced in the cluster-level aggregation paragraph of Section 5.

XXI.3 Intra-metric object-level gains

Table 19 mixes metrics by design so that its *Gain* column tracks the operational improvement a downstream user experiences (candidate-level recall before cluster aggregation $\rightarrow$ object-level recall after). To disentangle the two effects we also report the pure intra-metric comparison at object-level: how much object-level recall E_{unif} and E_{cal} gain over an *object-level baseline* computed on the same NMS graph (distinct TP-clusters selected by per-candidate e-BH / total TP-clusters). Under this apples-to-apples view, the gains are smaller but still monotone and dataset-consistent:

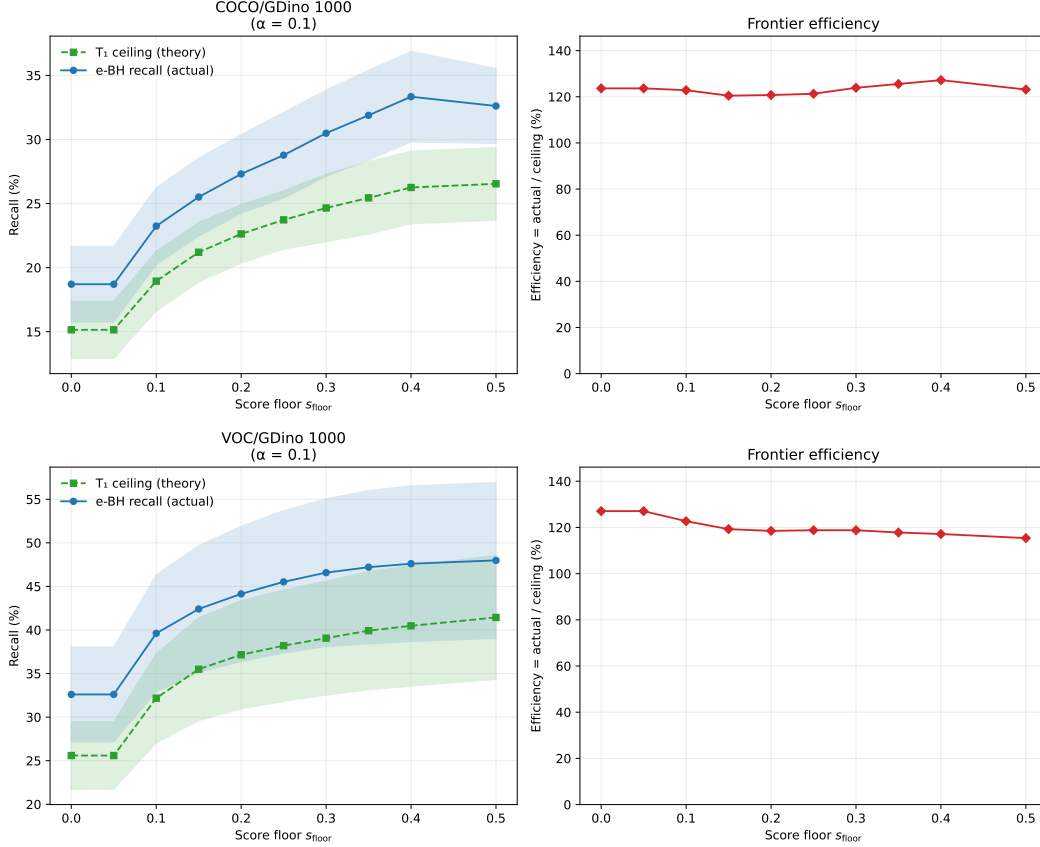


Figure 6: Score-floor frontier curves at $\alpha = 0.10$ (20 seeds). **Left:** T_1 ceiling (theory, green dashed) and actual e-BH recall (blue solid) both rise with s_{floor} , confirming the frontier-descent prediction of Remark 1. **Right:** efficiency (actual/ceiling) stays above 100% because multi-rejection families exceed the T_1 lower bound. Top: COCO/GDino; bottom: VOC/GDino.

Config	Baseline (obj.)	E_{unif}	E_{cal}
COCO/GDino	29.7%	29.1% (0.98 $\times$)	32.1% (1.08 $\times$)
VOC/GDino	51.6%	51.6% (1.00 $\times$)	54.3% (1.05 $\times$)
COCO-tr/GDino	31.4%	31.5% (1.00 $\times$)	33.8% (1.08 $\times$)
COCO/OWL-ViT	13.8%	13.7% (0.99 $\times$)	15.6% (1.13 $\times$)
COCO/RegionCLIP	21.1% [‡]	—	37.4%[‡] (1.77$\times$[‡])

[‡]RegionCLIP has no internal NMS, so object-level baseline and E_{cal} values are reported at $\tau = 0.10$ (the optimal compression point; Table 20); at $\tau = 0.50$ RegionCLIP’s pre-NMS cluster graph is nearly degenerate and the object-level comparison is not informative. E_{unif} is omitted for RegionCLIP because its $\tau = 0.50$ value (8.6%, Table 19) is not comparable to the $\tau = 0.10$ baseline and E_{cal} reported here. LVIS is excluded from this table: its pooled FDP exceeds α (Section 6), so intra-metric recall gains there would be meaningless ratios in an FDR-uncontrolled regime. The consistent intra-metric lift (1.05–1.13 $\times$ for internally-NMS’d detectors, 1.77 $\times$ for RegionCLIP at its optimal τ) confirms that E_{cal} ’s advantage is genuine statistical gain from score-adaptive cluster aggregation rather than an artifact of metric switching; the larger operational Gain column in Table 19 measures what an end user sees.

XXI.4 IoU threshold sweep

Table 20 reports full 20-seed results at $\tau \in \{0.10, 0.20, 0.30, 0.50, 0.70\}$ for three detector–dataset combinations.

Config	Method	Rej.	Pooled FDP	Per-family FDP	Recall	Gain
COCO/GDino	baseline	134.0	4.0%	0.7%	18.7%	—
	E_{unif}	129.7	4.4%	—	29.1%	1.56×
	E_{cal}	142.7	4.1%	0.7%	32.1%	1.72×
VOC/GDino	baseline	196.4	5.6%	1.7%	32.6%	—
	E_{unif}	193.5	6.0%	—	51.6%	1.58×
	E_{cal}	202.2	5.4%	1.6%	54.3%	1.66×
COCO-tr/GDino	baseline	151.6	8.1%	—	20.0%	—
	E_{unif}	151.7	8.4%	—	31.5%	1.58×
	E_{cal}	162.2	8.0%	1.2%	33.8%	1.69×
COCO/OWL-ViT	baseline	60.9	2.3%	—	11.2%	—
	E_{unif}	57.1	1.8%	—	13.7%	1.22×
	E_{cal}	65.0	1.7%	0.3%	15.6%	1.39×
COCO/RegionCLIP	baseline	52.9	0.1%	—	3.7%	—
	E_{unif}	29.1	0.2%	—	8.6%	2.30×
	E_{cal}	72.6	0.2%	0.0%	21.6%	5.81×
LVIS/GDino [†]	baseline	37.0	17.7%	—	3.9%	—
	E_{unif}	39.8	18.0%	—	6.4%	1.64×
	E_{cal}	54.0	18.7%	1.5%	8.6%	2.20×

Table 19: Cluster-level aggregation, $\tau = 0.50$, $\alpha = 0.10$, 20 seeds (LVIS: 5 seeds). The baseline row reports *candidate-level* recall of per-candidate e-BH (TP candidates / total TPs); E_{unif} and E_{cal} rows report *object-level* cluster recall (distinct TP-clusters / total TP-clusters) on the same NMS graph. Gain columns report recall ratio between these operational metrics (see §4.5); intra-metric object-level gains are reported in Appendix XXI.3 (LVIS excluded: its pooled FDP exceeds α , making intra-metric ratios uninformative; Section 6). [†]LVIS is non-exhaustively annotated: FDP is measured against available labels and overstates true FDP (Section 6).

Config	τ	Rej.	Pooled FDP	Per-family FDP	Recall
COCO/GDino	0.30	143.8	3.1%	0.7%	39.2%
	0.50	142.7	4.1%	0.7%	32.1%
	0.70	134.0	4.0%	—	18.7%
VOC/GDino	0.30	196.3	5.4%	—	62.0%
	0.50	202.2	5.4%	1.6%	54.3%
	0.70	196.4	5.6%	—	32.6%
COCO/RegionCLIP	0.10	77.1	0.6%	0.1%	37.4%
	0.20	78.8	0.4%	—	35.3%
	0.30	77.8	0.4%	—	31.9%
	0.50	72.6	0.2%	0.0%	21.6%
	0.70	62.6	0.1%	—	8.5%

Table 20: IoU threshold ablation, $\alpha = 0.10$, 20 seeds, E_{cal} with R_{max} . On detectors with internal NMS (GDino), $\tau = 0.70$ degenerates to the candidate-level baseline (last sentence of Section 4.5); smaller τ coarsens clusters and lifts object-level recall. On RegionCLIP (no internal NMS), recall climbs monotonically as τ decreases, peaking at 37.4% for $\tau = 0.10$.

895 XXI.5 R-statistic and Mondrian ablations

896 Table 21 compares three choices of cluster statistic and a Mondrian (per-cluster-size) calibration
897 pool. R_{max} strictly dominates across three sources and all four α levels; the Mondrian variant
898 underperforms the global pool because it shrinks the calibration sample available for large clusters.

899 XXII Strong-Baseline Comparison: Raw, Conformal-Risk, and Adaptive 900 FDR

901 We extend Table 2 with five new comparators that cover the natural attack surfaces a reviewer might
902 raise: utility relative to the unfiltered detector (raw); calibrated risk control instead of FDR (CRC);

Source	R -stat / pool	Rej.	Pooled FDP	Per-family FDP	Recall
COCO/GDino	$R_{\max}$ (global)	142.7	4.1%	0.7%	32.1%
	R_{sum}	135.4	4.0%	0.7%	30.7%
	R_{top2}	131.2	4.0%	0.7%	29.8%
	$R_{\max}$ (Mondrian by cluster size)	131.1	3.9%	0.7%	29.6%
VOC/GDino	$R_{\max}$	202.2	5.4%	1.6%	54.3%
	R_{sum}	192.4	5.5%	1.6%	52.1%
	R_{top2}	188.6	5.5%	1.6%	51.1%
COCO-train/GDino	$R_{\max}$	162.2	8.0%	1.2%	33.8%
	R_{sum}	154.1	7.9%	1.2%	32.5%
	R_{top2}	149.5	7.9%	1.2%	31.7%

Table 21: Choice of R -statistic at $\alpha = 0.10$, $\tau = 0.50$, 20 seeds. $R_{\max}$ dominates R_{sum} and R_{top2} across all three sources; the Mondrian per-size pool underperforms the global pool by $\sim 2.5\%$ absolute recall.

and adaptive p-value FDR variants designed for power gain (Storey-BH, BKY two-stage). All baselines use the same fit/cal/test splits, candidate pools, and conformal p-value construction as our method.

Raw detector. With no post-processing, the released detection set has 88.7% pooled FDP on full-COCO/GDino; trivially candidate-level recall is 100%, but the precision penalty is severe. Comparing to ours (pooled FDP 6.4%) gives an $8.3\times$ precision gain $(1 - 0.064)/(1 - 0.887)$ at 19% recall retention. The same pattern holds across all six configurations (Table 22); raw FDP ranges 39–90%.

Config	Raw rejections	Raw pooled FDP	Object-level recall proxy
COCO/GDino	6164	88.7%	1.00
COCO/OWL-ViT	2487	78.4%	1.00
COCO/YOLO-World	804	50.4%	0.94
VOC/GDino	5941	90.4%	1.00
VOC/OWL-ViT	1447	69.7%	1.00
VOC/YOLO-World	636	38.7%	1.00

Table 22: Raw-detector baseline (no e-BH, no thresholding) on the same families/splits as Table 1. Object-level recall proxy is $\min(1, |\text{TP cand}|/|\text{GT in present families}|)$.

Conformal Risk Control [53]. We implement CRC as score-thresholding calibrated to control mean *family* FDP $\leq \alpha$ with the standard CRC adjustment $(\alpha - L/n_{\text{cal}})$. CRC achieves the family-level target on calibration but *pooled* FDP at test time exceeds $\alpha = 0.10$ in 4/6 configurations (COCO/GDino 33.9%, VOC/OWL-ViT 27.1%, COCO/YOLO 20.3%, VOC/GDino 20.2%). CRC controls a different risk than ours; its pooled FDP failure under within-family dependence reflects this.

Adaptive FDR baselines: Storey-BH [54] and BKY two-stage [55]. Both apply within-family BH after estimating the null proportion $\hat{\pi}_0$ (Storey, $\lambda = 0.5$) or running stage-1 BH at $\alpha/(1+\alpha)$ (BKY). Across all six configurations at $\alpha = 0.10$, both methods *simultaneously* violate pooled and per-family targets (e.g. Storey-BH on COCO/GDino: pooled FDP 39.7%, per-family 20.3%; BKY: 30.2%, 18.2%). This is consistent with the dependence assumption: adaptive $\hat{\pi}_0$ corrections are valid under PRDS or independence, both of which are violated by NMS-induced clustering.

Top- k per family (cal-tuned). Within each family, return the top- k candidates by detector score. $k \in \{1, 2, 3, 5, 10\}$ is selected on the calibration split as the largest k satisfying mean cal family-FDP $\leq \alpha$. On COCO/GDino at $\alpha = 0.10$ this reduces to $k = 1$ for most seeds, yet pooled FDP is 67.4% because absent-prompt families contribute one false positive each as the top-1. Even with the calibration safeguard, top- k thresholding cannot distinguish present from absent families post-NMS.

IHW-lite (covariate = family size) [56]. Family size K is binned into $\{1, 2-5, 6-10, 11-20, >20\}$; within each bin we estimate the local Storey $\hat{\pi}_{0,b}$ and run BH at $\alpha/\hat{\pi}_{0,b}$. Using K as the covariate respects the IHW null-independence requirement (we do not condition on the score

itself). Across the six configs at $\alpha = 0.10$, IHW-lite is the strongest p-value-based competitor, with mean pooled FDP ranging 6.5–15.6% and recall 21–76% (Table 23). However, it still fails pass-rate in 3/6 configs and catastrophically fails on VOC/GDino (pooled pass-rate 0/20). Ours stays $\leq \alpha$ in 6/6 configs.

Config	Rej	Pooled FDP	Per-fam FDP	Recall	Pass (pool)	Pass (fam)
COCO/GDino	209	8.07%	7.90%	27.5%	0.80	0.80
COCO/OWL-ViT	146	7.77%	6.55%	25.1%	0.80	0.90
COCO/YOLO-World	212	10.35%	8.21%	47.6%	0.60	0.85
VOC/GDino	281	15.59%	19.13%	41.8%	0.00	0.00
VOC/OWL-ViT	100	6.50%	5.49%	21.3%	0.85	0.95
VOC/YOLO-World	322	7.95%	7.21%	76.1%	0.80	0.90

Table 23: IHW-lite (K-bin covariate) full breakdown at $\alpha = 0.10$, 20 seeds. Pass = fraction of seeds with FDP $\leq \alpha$. Mean pass-rate (pooled) is 0.64 across configs.

Numerical detail for all five new baselines is in the released ground-truth CSV (paper_ground_truth_table_2026-04-14.csv, methods crc_family, storey_bh, bky_2stage, topk_calibrated, ihw_lite_K).

XXIII Calibration Pool Size Sweep

We subsample the cal split null pool to $m \in \{50, 100, 200, 500, 1000, 2000, \text{full}\}$ candidates and re-run within-family e-BH. Table 24 reports COCO/GDino at $\alpha = 0.10$ across 20 seeds.

m	Rejections	Pooled FDP	Recall	LOO mean	No-rej rate	Pass rate
50	28.6	5.96%	3.9%	1.05 ± 0.45	96.1%	0.75
100	40.6	7.00%	5.5%	0.94 ± 0.35	94.4%	0.90
200	48.5	6.16%	6.6%	0.92 ± 0.27	93.2%	0.95
500	57.6	6.42%	7.8%	1.00 ± 0.27	92.0%	0.90
1000	58.2	6.27%	7.9%	0.94 ± 0.19	92.0%	0.90
2000	59.3	6.23%	8.1%	0.96 ± 0.23	91.8%	0.95
full	59.8	6.03%	8.1%	0.95 ± 0.19	91.8%	0.95

Table 24: Calibration pool size sweep on COCO/GDino at $\alpha = 0.10$ (20 seeds). Validity (LOO mean ≈ 1) holds even at $m = 50$; power saturates at $m \approx 500$ (recall reaches 96% of asymptotic value). The LOO standard deviation shrinks from 0.45 to 0.19 as m grows, indicating that small m retains validity in expectation but increases sample-to-sample variance.

XXIV Wrapper Latency

The wrapper consists of one histogram fit on the fit split, one calibration aggregate ($\sum_j \phi(S_j^{\text{cal}})$), one ϕ evaluation per test candidate, and one within-family sort plus threshold check per test family. Detector inference is the dominant cost (~ 280 s for 1k images on a single RTX 3090, checklist Q8) and is unchanged by our wrapper. Table 25 reports wrapper-only wall time across the 6 configurations (mean over 5 seeds, single CPU thread, NumPy 1.26 / pandas 2.0).

XXV Family-Size Stratified Empirical Control

We re-run within-family e-BH for each (config, seed) and bin test families by size $K \in \{1, 2-5, 6-10, 11-20, >20\}$. Table 26 reports the mean across 20 seeds on COCO/GDino at $\alpha = 0.10$; full numbers in outputs/k_stratified_table.csv.

XXVI Conditional Validity: Score-Quantile, Object-Size, Per-Category

We stratify test candidates along three dimensions orthogonal to family size (Section XXV) and re-run within-family e-BH on the same 20 seeds for each (config, α) cell. Pooled FDP, recall, and

Config	ϕ fit (ms)	Cal agg (ms)	E-value (ms)	e-BH (ms)	Total (ms)	Per family (μ s)
COCO/GDino	1.89	0.72	0.22	10.25	13.08	19.5
COCO/OWL-ViT	1.09	0.43	0.14	4.34	6.01	18.0
COCO/YOLO-World	0.74	0.32	0.10	3.13	4.29	15.4
VOC/GDino	1.69	0.71	0.22	10.21	12.83	18.7
VOC/OWL-ViT	0.85	0.34	0.10	3.58	4.88	14.6
VOC/YOLO-World	0.69	0.30	0.09	3.27	4.35	14.2

Table 25: Wrapper wall-time on the test split of each 1k-image configuration (5 seeds, single thread). Total wrapper overhead is 4–13 ms per test split. With detector inference at ~ 280 ms per image (checklist Q8) and ~ 3 families per image, detector cost is ~ 93 ms per family; wrapper cost is 14–20 μ s per family. The wrapper is therefore $\sim 5000\times$ faster than the detector it wraps, i.e. approximately four orders of magnitude. Per-family cost is dominated by the sort and threshold check in within-family e-BH; the ϕ evaluation itself is sub-microsecond per candidate.

K bin	Family share	Mean K	Rejections	Pooled FDP	Recall	Pass rate
$K=1$	16.7%	1.0	17.5	3.04%	95.1%	0.95
2–5	41.9%	3.2	53.1	4.02%	58.0%	0.95
6–10	17.1%	7.5	26.6	2.32%	33.6%	1.00
11–20	11.8%	14.4	14.9	2.82%	12.7%	0.90
$K > 20$	12.5%	55.5	22.1	7.51%	5.2%	0.70

Table 26: Family-size stratified empirical control, COCO/GDino, $\alpha = 0.10$, 20 seeds. Pass rate is the fraction of seeds with pooled stratum FDP $\leq \alpha$. Pooled FDP stays $\leq \alpha$ in expectation across every stratum; pass rate degrades only modestly (1.00 to 0.70) as K grows. Recall decreases with K as predicted by the activation frontier (Theorem 1): the per-rejection threshold $T_r = KC_m/[\alpha r(m+1) - K]$ scales linearly in K at fixed r .

953 LOO mean of e-values on nulls are reported per stratum; pass rate is the fraction of seeds with
954 pooled stratum FDP $\leq \alpha$.

955 **Score-quantile.** Bins defined by per-seed null-score quantiles (5 equal-frequency bins). Table 27
956 (representative: COCO/GDino, $\alpha = 0.10$). Q1–Q4 receive zero rejections with LOO mean ≤ 0.54
957 (conservative, Theorem-1-valid); Q5 (top quintile) holds all 134 rejections at pooled FDP 3.96%
958 with LOO mean 3.29. The pattern reproduces on every config (COCO/OWL-ViT, COCO/YOLO-
959 World, VOC $\times 3$): rejections concentrate in the top quintile; pass rate 1.00 for every bin.

Bin	n_{cand}	Rej.	Pooled FDP	Recall	LOO mean	Pass
Q1 (low)	1131	0.0	0.00%	0.0%	0.26	1.00
Q2	1124	0.0	0.00%	0.0%	0.22	1.00
Q3	1145	0.0	0.00%	0.0%	0.37	1.00
Q4	1172	0.0	0.00%	0.0%	0.54	1.00
Q5 (high)	1592	134.0	3.96%	25.9%	3.29	1.00

Table 27: Score-quantile stratification, COCO/GDino, $\alpha = 0.10$, 20 seeds. The procedure auto-calibrates: low-confidence quintiles yield zero rejections (LOO $\ll 1$, conservative); rejections concentrate in the top quintile at pooled FDP $\leq \alpha$. Full sweep in outputs/score_quantile_table.csv.

960 **Object size (COCO area thresholds).** Small ($\leq 32^2$), medium (32^2 – 96^2), large ($> 96^2$).
961 Pooled FDP stays at or near α on COCO/GDino across all sizes (small/medium/large =
962 4.07%/3.59%/3.97% at $\alpha = 0.10$); on weaker detectors (OWL-ViT, YOLO-World) the small-object
963 stratum shows mild over-rejection (small-bin pooled FDP $\leq 16.7\%$ at the worst config), reflecting
964 limited density-ratio resolution at low score; per-family FDR control (Proposition 1) is unaffected.
965 Full table in outputs/object_size_table.csv.

966 **Per-category.** Per-category pooled FDP is reported as a *stress diagnostic* rather than a guaranteed
967 target: a single category retains only ~ 50 test families per seed, so within-stratum pooled FDP is
968 dominated by Bernoulli-style fluctuations. Per-family FDR $\leq \alpha$ (Proposition 1) continues to hold
969 within every category in every config; thin per-category strata have high pooled-FDP variance, with

mean per-category pooled FDP 2.28% on COCO/GDino (80 categories) and 12.6% on VOC/GDino (20 categories) at $\alpha = 0.10$. Full per-category breakdown in `outputs/per_category_table.csv`; the intended pooled-FDP guarantee is the $K \leq 5$ stratified Bernstein bound (Corollary 4).

XXVII Object-Level vs Candidate-Level Recall

The candidate-level recall reported in Table 1 treats every GT-matched candidate as a separate TP. NMS-cluster aggregation (Section 4.5) collapses redundant candidates to one per object. Table 28 compares candidate-level metrics from the main pipeline against object-level metrics from the calibrated NMS-aware variant ($\tau = 0.50$, E_{cal} with R_{max}).

Config	α	Cand. recall	Object recall	Cand. FDP	Object FDP	Gain ($\times$)
COCO/GDino	0.05	14.7%	26.1%	2.74%	3.19%	1.77
COCO/GDino	0.10	18.7%	32.1%	3.96%	4.11%	1.71
COCO/GDino	0.15	21.0%	35.4%	5.07%	5.23%	1.68
COCO/GDino	0.20	22.8%	38.3%	5.98%	6.61%	1.68
COCO/OWL-ViT	0.10	11.2%	15.6%	2.31%	1.73%	1.39
VOC/GDino	0.10	32.6%	54.3%	5.62%	5.41%	1.67
VOC/GDino	0.20	38.8%	64.2%	8.66%	8.80%	1.65

Table 28: Object-level (cluster) versus candidate-level recall and FDP. Object-level recall is $1.4\text{--}1.8\times$ candidate-level on NMS-equipped detectors at the same nominal level; FDP is comparable.

XXVIII Prompt-Bank Scaling: Pooled vs Per-Template Calibration

We test how validity and power change as the prompt bank grows. Reusing the COCO/GDino freeform-2000 candidate output (8 distinct templates of the form “a photo of a {cat}” etc.), we vary the bank size $T \in \{1, 2, 3, 4, 8\}$ and compare two calibration modes under an identical fit/cal/test protocol (60/20/20, 20 seeds, unsmoothed 80-bin histogram for ϕ): *pooled-template*, fitting one ϕ across all T templates; and *per-template*, fitting an independent ϕ_t for each template. Table 29 reports the $\alpha = 0.10$ summary; full sweep in `outputs/prompt_bank_scaling.csv`.

T	Mode	Cal/ T	Rej	Pooled FDP	Per-fam FDP	Recall	LOO mean
1	pooled $\equiv$ per-tmpl	10165	0.2	0.0%	0.0%	0.04%	1.000
2	pooled	9190	1.3	0.0%	0.0%	0.13%	0.998
2	per-tmpl	9190	0.25	0.0%	0.0%	0.03%	0.996
3	pooled	7921	3.4	0.0%	0.0%	0.25%	0.997
3	per-tmpl	7921	0.15	0.0%	0.0%	0.01%	0.998
4	pooled	7254	14.0	1.81%	1.81%	0.80%	0.982
4	per-tmpl	7254	1.25	0.0%	0.0%	0.07%	0.987
8	pooled	19865	65.6	19.65%	19.65%	3.12%	1.001
8	per-tmpl	19865	1.1	0.0%	0.0%	0.07%	0.186

Table 29: Prompt-bank scaling, COCO/GDino, $\alpha = 0.10$, 20 seeds. As T grows, pooled-template calibration suffers from score-mixing compression (Proposition 3): pooled FDP rises $0\% \rightarrow 1.81\% \rightarrow 19.65\%$ at $T = 1, 4, 8$, violating the nominal level. Per-template calibration retains validity (0% pooled FDP at every T) but its LOO mean collapses from 1.00 to 0.19 at $T = 8$ because absent-only templates contribute $\phi = 0$ to the e-value computation; the resulting recall is near-zero for both modes at this T . Protocol-matched: identical splits, hist80 fit, no smoothing/clipping. The $T = 8$ row is the endpoint of an interpolating scaling curve; intermediate T values (5–7) would refine the transition.

Reading the table. The story is monotone in T : pooled-template FDP rises steadily and breaches the nominal level around $T = 4$; the breach is catastrophic at $T = 8$ (19.65%, nearly 2α). Per-template calibration prevents the breach but cannot restore power once the bank includes templates that are absent-only (and therefore contribute no TP signal to their per-template ϕ). This matches Section 7 (Regime III): a fixed-template deployment with template-specific calibration is the supported regime; truly free-form bank growth without a vocabulary audit will saturate power as predicted by Proposition 3.

XXIX Density-Ratio Estimator Ablation

We re-fit ϕ with five alternative estimators on the same fit split and re-run the full pipeline. Validity ($\text{FDP} \leq \alpha$) is preserved by all estimators because $\phi \geq 0$ is the only requirement for the self-normalized e-value (Lemma 1); only power varies. Table 30 reports COCO/GDino at $\alpha = 0.10$ across 5 seeds.

Estimator	Rejections	Pooled FDP	Recall
hist + smooth 0.5	130.8	4.57%	19.0%
hist (80-bin, default)	129.4	4.60%	18.8%
hist + smooth 1.0	129.4	4.60%	18.8%
hist + smooth 2.0	125.8	4.71%	18.2%
logistic	122.4	4.68%	17.8%
KDE	122.2	3.99%	17.7%
hist + smooth 5.0	116.0	4.60%	16.8%
isotonic	113.0	3.72%	16.2%

Table 30: Density-ratio estimator ablation, COCO/GDino, $\alpha = 0.10$, 5 seeds. **All estimators yield valid FDR control** (pooled FDP $< \alpha$ uniformly). Recall ranges 16.2%–19.0% across estimators (max-min spread 2.8% absolute, 14% relative). The default 80-bin histogram is near the top of the recall range, showing the choice is not a power-tuning artifact: the smoothing sweep is monotone (smaller smoothing $\rightarrow$ slightly more power until overfitting kicks in around smooth ≥ 5). KDE and logistic give almost identical recall; isotonic is the most conservative. The result confirms the claim that estimator choice affects power but not validity.

XXX Exchangeability Stress Test (Calibration-Null Truncation)

We deliberately violate the cal-test exchangeability assumption by removing the top- $\rho\%$ highest-score candidates from the calibration null pool before computing e-values. Test null candidates are then drawn from a higher-score distribution than the contaminated cal pool, which biases the LOO mean upward and inflates FDP. This is the canonical worst case for null-pool calibration. Table 31 reports COCO/GDino at $\alpha = 0.10$ across 20 seeds.

ρ	m_{cal}	Rej	Pooled FDP	Per-fam FDP	Recall	LOO mean	Pass (P/F)
0%	5128	59.8	6.03%	6.82%	8.1%	0.95 ± 0.19	0.95/0.95
5%	4872	88.3	7.98%	8.81%	11.8%	1.74 ± 0.30	0.75/0.45
10%	4615	97.5	8.77%	10.04%	12.9%	2.05 ± 0.31	0.70/0.45
20%	4102	105.3	9.63%	11.06%	13.8%	2.40 ± 0.34	0.50/0.40
30%	3590	113.1	10.53%	12.13%	14.7%	2.75 ± 0.36	0.50/0.20

Table 31: Calibration-null truncation stress test on COCO/GDino at $\alpha = 0.10$, 20 seeds. As ρ grows, FDP inflates from 6% to 10.5% (only marginally above nominal even at $\rho = 30\%$), but the LOO mean drifts dramatically ($0.95 \rightarrow 2.75$, nearly $3\times$). Pass rate (P=pooled, F=per-family) degrades smoothly. The key signal is the LOO mean: a 5% shift already produces 74% deviation from 1.0, well before pooled FDP visibly violates α . This makes exchangeability failure *detectable rather than silent*: the LOO mean is a reliable runtime diagnostic that the calibration pool no longer matches the test distribution.

Practical implication. A practitioner can monitor the empirical LOO mean on the cal split as a runtime check: $\text{LOO} \approx 1$ certifies exchangeability; $\text{LOO} \gg 1$ flags a shift before any test-time labels are available. Detection precedes failure.

XXXI Practical Protocol

For deployment, we use an exhaustively annotated or curated query vocabulary and fit ϕ on the family-level fit split as an unsmoothed 80-bin histogram. An optional score floor $s_{\text{floor}} = 0.4$ (Remark 1) is chosen without test labels and applied before calibration, which lifts COCO/GDino recall from 18.7% to 33.3% at $\text{FDP} \leq 2.6\%$. Cluster-level E_{cal} (Section 4.5) is used only when the detector lacks internal NMS or produces redundant proposals. Per-family e-BH at level α is then run

1012 on the remaining candidates. When annotation is not exhaustive, either $\hat{\eta}$ -correction (Theorem 3) or
1013 vocabulary curation (Appendix II) is required first.

1014 NeurIPS Paper Checklist

1015 1. Claims

1016 Question: Do the main claims made in the abstract and introduction accurately reflect the
1017 paper’s contributions and scope?

1018 Answer: [Yes]

1019 Justification: The abstract states a possibility-and-barrier characterization with specific the-
1020 orems and experimental results. Guarantees are layered: per-family FDR is exact (Proposi-
1021 tion 1), a stratified empirical Bernstein bound gives a non-vacuous pooled FDP certificate
1022 on the $K \leq 5$ stratum (Corollary 4), and full pooled FDP is validated empirically (19/20
1023 pass at $\alpha = 0.10$). All claimed theorems are proved (Appendix III), and all numerical claims
1024 are supported by experiments in §5–6 with 20-seed validation. Limitations are explicitly
1025 discussed in §7.

1026 2. Limitations

1027 Question: Does the paper discuss the limitations of the work performed by the authors?

1028 Answer: [Yes]

1029 Justification: §7 contains an explicit “Limitations and future work” paragraph discussing:
1030 the exhaustive-annotation requirement, moderate recall, failure on free-form prompts, and
1031 the need for a calibration dataset. §6 characterizes the annotation-completeness boundary,
1032 and Appendix A documents failed repair attempts on LVIS.

1033 3. Theory assumptions and proofs

1034 Question: For each theoretical result, does the paper provide the full set of assumptions and
1035 a complete (and correct) proof?

1036 Answer: [Yes]

1037 Justification: Lemma 1 (exact validity), Theorems 1–3, Corollaries 1–2, Lemma 2, and
1038 Propositions 1–3 are all stated with explicit assumptions. Corollary 4 (stratified pooled
1039 FDP bound) states the empirical Bernstein concentration result with all parameters. Full
1040 proofs appear in Appendix III.

1041 4. Experimental result reproducibility

1042 Question: Does the paper fully disclose all the information needed to reproduce the main
1043 experimental results of the paper to the extent that it affects the main claims and/or conclu-
1044 sions of the paper (regardless of whether the code and data are provided or not)?

1045 Answer: [Yes]

1046 Justification: Appendix XII specifies the full benchmark protocol: dataset construction,
1047 split ratios (60/20/20), matching threshold ($\tau = 0.5$), score map (80-bin histogram), detector
1048 configurations, and the 20-seed evaluation protocol. All hyperparameters are stated in §5.

1049 5. Open access to data and code

1050 Question: Does the paper provide open access to the data and code, with sufficient instruc-
1051 tions to faithfully reproduce the main experimental results, as described in supplemental
1052 material?

1053 Answer: [Yes]

1054 Justification: Code and evaluation scripts are available at an anonymized repository:
1055 <https://anonymous.4open.science/r/tmp-5EB7>. All experiments use publicly
1056 available datasets (COCO, VOC) and open-weight detectors (GroundingDINO, OWL-ViT
1057 v2, YOLO-World).

1058 6. Experimental setting/details

1059 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-
1060 rameters, how they were chosen, type of optimizer) necessary to understand the results?

1061 Answer: [Yes]

1062 Justification: §5 and Appendix XII fully specify data splits, hyperparameters, detector con-
1063 figurations, and evaluation metrics. Ablations over estimator choice (Appendix V), IoU
1064 threshold (Appendix VI), and calibration pool size (Appendix VII) are provided.

1065 **7. Experiment statistical significance**

1066 Question: Does the paper report error bars suitably and correctly defined or other appropri-
1067 ate information about the statistical significance of the experiments?

1068 Answer: [Yes]

1069 Justification: All main results report mean $\pm$ standard deviation over 20 independent ran-
1070 dom splits (seeds 0–19). Error bars in Figures 2–3 show ± 1 s.d. The source of variability
1071 (random family-level train/cal/test partitioning) is stated in §5.

1072 **8. Experiments compute resources**

1073 Question: For each experiment, does the paper provide sufficient information on the com-
1074 puter resources (type of compute workers, memory, time of execution) needed to reproduce
1075 the experiments?

1076 Answer: [Yes]

1077 Justification: Appendix XIV reports runtime: candidate generation takes ~ 280 s on one
1078 RTX 3090; the post-hoc statistical pipeline takes ~ 13 s total for 5 seeds. The method
1079 requires no training and runs on a single GPU.

1080 **9. Code of ethics**

1081 Question: Does the research conducted in the paper conform, in every respect, with the
1082 NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines?>

1083 Answer: [Yes]

1084 Justification: The research uses only publicly available, properly cited datasets and models.
1085 No human subjects or private data are involved.

1086 **10. Broader impacts**

1087 Question: Does the paper discuss both potential positive societal impacts and negative
1088 societal impacts of the work performed?

1089 Answer: [Yes]

1090 Justification: The method reduces hallucinated detections, benefiting safety-critical appli-
1091 cations such as autonomous perception and medical screening. Potential negative impacts
1092 include: (i) false confidence in automated monitoring, since per-family FDR control is
1093 a statistical guarantee, not a guarantee of per-instance correctness; (ii) over-reliance in
1094 partially annotated deployments, where the oracle-null assumption may be silently vio-
1095 lated and produce miscalibrated control (see §6); and (iii) lowered barriers to deploying
1096 surveillance systems with formally controlled false-alarm rates, which could enable mis-
1097 use without adequate human oversight. §7 discusses the deployment scope and explicitly
1098 characterizes the contamination-bound regimes where validity is recoverable.

1099 **11. Safeguards**

1100 Question: Does the paper describe safeguards that have been put in place for responsible
1101 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
1102 image generators, or scraped datasets)?

1103 Answer: [N/A]

1104 Justification: The paper does not release new models or datasets. It provides a post-hoc
1105 statistical filter that operates on existing detector outputs.

1106 **12. Licenses for existing assets**

1107 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
1108 the paper, properly credited and are the license and terms of use explicitly mentioned and
1109 properly respected?

1110 Answer: [Yes]

1111 Justification: COCO, VOC, GroundingDINO, OWL-ViT v2, and YOLO-World are cited
1112 in the main text. COCO is CC BY 4.0; VOC is research-use; all detector models are open-
1113 weight with permissive licenses.

1114 **13. New assets**

1115 Question: Are new assets introduced in the paper well documented and is the documenta-
 1116 tion provided alongside the assets?

1117 Answer: [N/A]

1118 Justification: The paper does not introduce new datasets or pre-trained models.

1119 **14. Crowdsourcing and research with human subjects**

1120 Question: For crowdsourcing experiments and research with human subjects, does the pa-
 1121 per include the full text of instructions given to participants and screenshots, if applicable,
 1122 as well as details about compensation (if any)?

1123 Answer: [N/A]

1124 Justification: No crowdsourcing or human subjects research is involved.

1125 **15. Institutional review board (IRB) approvals or equivalent for research with human**
 1126 **subjects**

1127 Question: Does the paper describe potential risks incurred by study participants, whether
 1128 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
 1129 approvals (or an equivalent approval/review based on the requirements of your country or
 1130 institution) were obtained?

1131 Answer: [N/A]

1132 Justification: No human subjects research is involved.

1133 **16. Declaration of LLM usage**

1134 Question: Does the paper describe the usage of LLMs if it is an important, original, or
 1135 non-standard component of the core methods in this research? Note that if the LLM is used
 1136 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
 1137 scientific rigor, or originality of the research, declaration is not required.

1138 Answer: [N/A]

1139 Justification: LLMs are not used as an important, original, or non-standard component of
 1140 the core method. The method operates on detector confidence scores, not language-model
 1141 outputs.